

Radiation Interactions

9.1 Introduction

Early in the space program, it was recognized that radiation was a potentially serious hazard to both astronauts and electronics. On Earth, we are protected from radiation by shielding of the Earth that restricts potential radiation to a hemisphere above us, by the geomagnetic field that deflects charged particles, and by the Earth's atmosphere that dissipates some radiation. Radiation remains a critical consideration in the selection of electronic parts, especially semiconductors, and the exposure of humans especially in missions outside of the geomagnetic field. High-energy protons, electrons, and heavy ions lose energy as they interact with materials through non-ionizing and ionizing processes. Radiation that has enough energy to move atoms in a molecule or cause them to vibrate, but not remove electrons, is referred to as non-ionizing radiation. Examples of this kind of radiation are radio waves, microwaves, and visible light. *Non-ionizing radiation* can produce damage in material and tissue by the production of heat or atomic displacements. *Ionizing radiation* has sufficient energy to remove bound electrons from atoms, thus creating ions and charge distributions. Ionizing radiation can cause the malfunction of electronic devices, especially semiconductors, and damage tissue, causing a variety of radiation sicknesses including cancer.

Damage to electrical devices is generally categorized as non-ionizing thermal damage, displacement damage, total ionization dose damage, and single event effects. *Non-ionizing thermal damage* results from an increase in temperature, which is a function of the energy density of the radiation. *Displacement damage* is the result of the creation of vacancies or interstitials in bulk material. *Total ionization dose damage* is a cumulative effect in which electrical parameters and performance of a device are changed from their pre-ionization values due to undesirable charge distribution. *Single event phenomena* are circuit changes, upsets, or damage caused by one or a small number of energetic photons or particles. Some electrical components are much less susceptible to radiation damage than others. Passive electrical components, such as resistors, capacitors, and inductors are relatively immune. Active electrical components such as field effect transistors, zener diodes, bipolar transistors, and integrated circuits are more readily affected. When a high-energy quantum of radiation impacts a tissue cell, it can cause no damage, cause damage that the cell can repair, cause damage that is passed on to its progeny, or cause the cell to die. Radiation effects on tissue are categorized into deterministic and stochastic effects.

Mitigating the effects of radiation depends on accurately assessing the radiation exposure, the radiation susceptibility of the material, and the level of acceptable risk. Determining exposure in space missions requires knowing the launch date, the mission duration, the trajectory, and mission scenario; models of the radiation environment; material properties of shielding and nearby structures; and the potential damage mechanisms. To minimize risk may require changing the trajectory and mission operations, employing radiation tolerant designs with radiation resistant parts, and adding shielding. In some cases, shielding may actually exacerbate the risk to devices or tissue by producing secondary radiation that may be more harmful than the primary radiation.

The sources of radiation in space of interest to spacecraft designers are onboard nuclear devices and the natural radiation environment that for the environs of the Earth consists of trapped radiation, solar wind, galactic cosmic rays, and solar particle events. In interplanetary missions, the radiation environment of the targeted celestial body is also important.

9.2 Radiation from Onboard Nuclear Devices

Nuclear devices are employed in space missions to produce electrical power, generate heat, and provide high specific impulses important for missions away from the Earth. The first radioisotope thermoelectric generator (RTG) used in a space mission was launched aboard a U.S. Navy Transit navigation satellite in 1961. The electrical power output was only 2.7 W but it continued to perform for 15 years after launch.

RTGs generate electrical power from the heat of radioactive decay of an isotope that is converted into electricity by thermocouples through either a thermoelectric or thermionic process. Typical thermoelectric efficiencies are about 7%, and thermionic efficiencies are about 12%. The general purpose heat source (GPHS) used today by NASA and illustrated in Fig. 9.1 uses plutonium-238 as its fuel and generates electricity thermoelectrically. A GPHS-RTG assembly weighs 56 kg, is approximately 113 cm long and 43 cm in diameter, and contains 10.9 kg of plutonium dioxide fuel. To minimize risk of a radioactive release, the fuel is stored in individual modular units, each with its own heat shield. A layer of iridium metal encased in high-strength graphite blocks surrounds each unit. These two materials are resistant to corrosion and heat. The GPHS is designed to be recovered intact from reentry into the Earth's atmosphere with near-Earth orbital velocities of $<8 \text{ km s}^{-1}$.

Radioisotope heater units (RHU) are used in some missions as an element of a spacecraft's thermal control systems, especially in missions to the outer planets where the sun's thermal energy is significantly reduced. The Cassini spacecraft and the Huygens Probe use approximately 117 RHUs. Each RHU provides about 1 W of heat, derived from the radioactive decay of 2.7 g of a nonweapons-grade radioisotope, plutonium-238 dioxide, contained in a platinum-30 rhodium alloy clad. An RHU has a diameter of 2.6 cm, length of 3.2 cm, and mass of 40 g, as illustrated in Fig. 9.2.

Plutonium-238 has a half-life of 87.7 years, and its radiation is principally alpha particles with low levels of neutron and gamma radiation. The primary

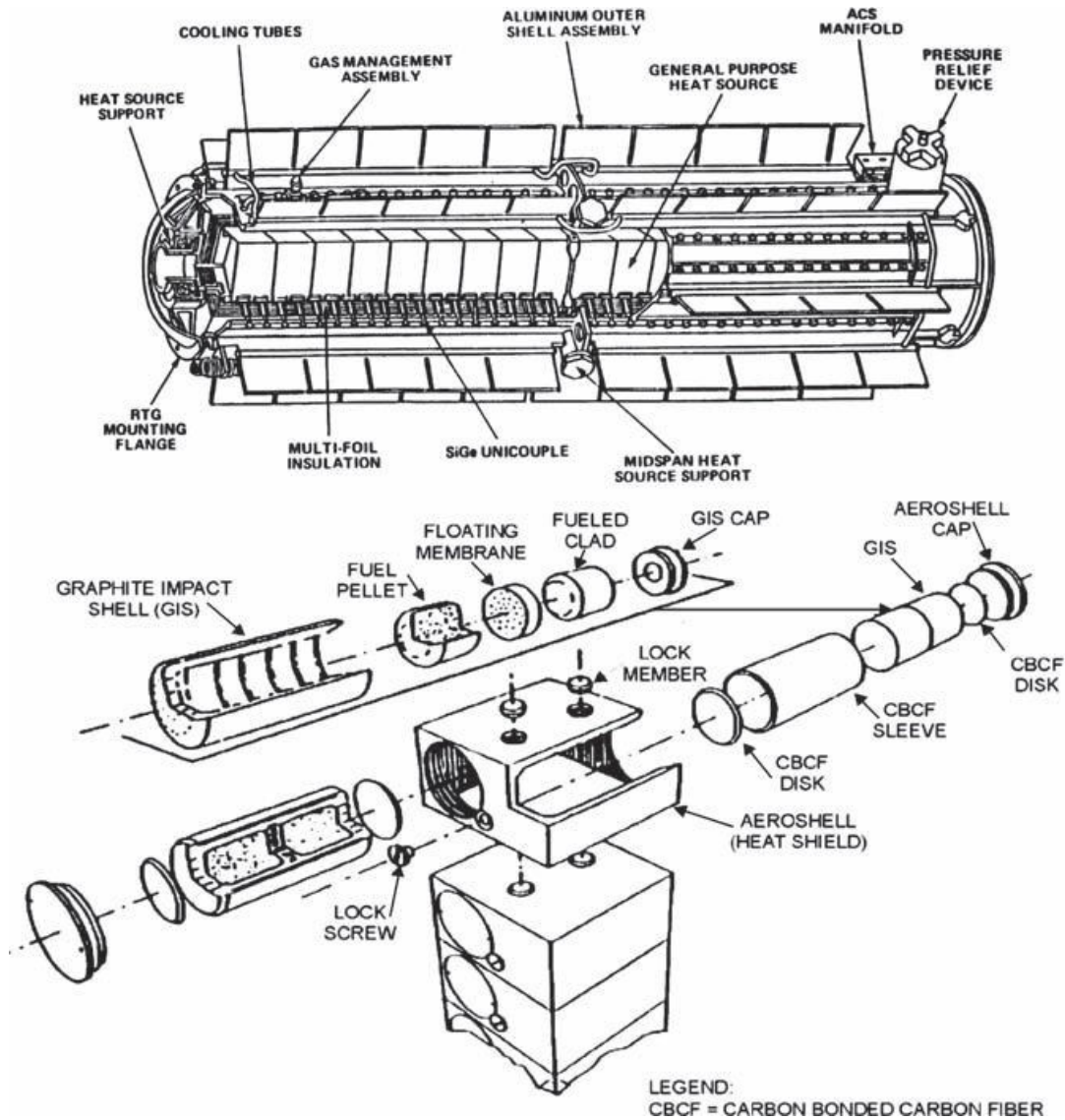


Fig. 9.1 General purpose heat source (GPHS). [Courtesy of U.S. Department of Energy (2007).]

human risk from plutonium is if small particles become airborne and are inhaled. An estimate of the acute toxic effect lethal dose is about 0.5 g, but the actual amount would depend on a variety of circumstances.

A nuclear reactor is a device in which nuclear chain reactions are initiated, controlled, and sustained at a steady rate. The United States launched the first space reactor, System for Nuclear Auxiliary Power (SNAP) 10A, in 1965. This small, 500-W electric, experimental reactor, like all subsequent space reactors, was fueled by uranium-235. It operated for 43 days until an apparent electrical malfunction. Between 1967 and 1988 the former Soviet Union launched 31 low-powered fission reactors with thermoelectric conversion in Radar Ocean Reconnaissance Satellites (RORSATs) missions. These were followed by the Topaz reactors with thermionic conversion that generated about 5 kW of electrical power.



Fig. 9.2 Radioisotope heater unit. [Courtesy of U.S. Department of Energy (2007).]

9.3 Natural Space Radiation Environment

The natural radiation environment is categorized into four types: solar wind, radiation trapped in a planetary magnetic field if it has one, galactic cosmic rays (GCR), and solar particle events (SPE), as discussed in Chapter 6.

Because of the extremely high temperature of the sun's corona, a steady stream of quanta of radiation (photons, protons, and electrons) escapes in a radial direction. This is one source of the constituents of the trapped radiation and is a component of the radiation to which an interplanetary mission is exposed. A celestial body with a magnetic field can trap particles, primarily electrons and protons, which result from collisions of high-energy cosmic rays with the planet's atmosphere and decelerations of the solar wind and solar particle events by magnetic storms. Galactic cosmic rays come from outside the solar system and are primarily atomic nuclei from which electrons have been stripped away. They have probably been accelerated over the last few million years, have traveled many times across the galaxy, and are trapped by the galactic magnetic field. Solar particle events are associated with coronal mass ejections that are more prevalent at higher solar activity. Correlation of the number density of the categories of radiation with solar activity is given in Table 6.4.

9.4 Photon Interactions

9.4.1 Introduction

Photon radiation is the transfer of energy by electromagnetic radiation that has zero rest mass and no charge. Forms of photon radiation include radio and television waves, microwaves, electric light, ultraviolet radiation, x-rays, and gamma rays. X-ray radiation is used to describe radiation emanating from atomic electrons and

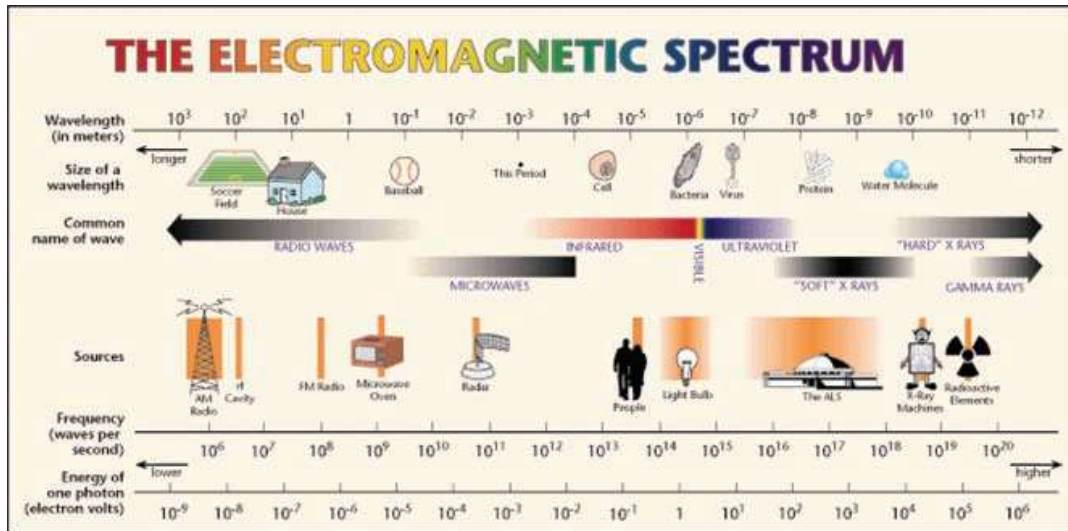


Fig. 9.3 The electromagnetic spectrum. [From Lawrence Berkeley National Laboratory (2007); courtesy of the Advanced Light Source, Berkeley Lab.]

typically has a frequency between 3×10^{16} and 3×10^{19} Hz, wavelength between 10^{-8} and 10^{-11} m, and quantum energy between 100 eV and 100 keV. Gamma radiation is used to denote electromagnetic radiation from the nucleus and typically has a frequency $>3 \times 10^{19}$ Hz, wavelength $<10^{-11}$ m, and quantum energy >100 keV. See Fig. 9.3 for the electromagnetic spectrum.

The relationship between energy, frequency, and wavelength is given by

$$E = hf \quad (9.1)$$

where

h = Planck's constant, J s or eV s

f = frequency, Hz

Example 9.1

Problem: Determine the frequency and wavelength of radiation with an energy of 100 keV.

Solution: The frequency for radiation with an energy of 100 keV is given by

$$f = E/h = (1 \times 10^5 \text{ eV}) / (4.14 \times 10^{-15} \text{ eV s}) = 2.42 \times 10^{19} \text{ Hz} \quad (9.2)$$

The wavelength can be determined from the frequency by

$$\lambda = cf^{-1} = 3 \times 10^8 \text{ m s}^{-1} / 2.42 \times 10^{19} \text{ cycles s}^{-1} = 1.24 \times 10^{-11} \text{ m cycle}^{-1} \quad (9.3)$$

where

c = speed of light, m s^{-1}

λ = wavelength, m

Important when dealing with radioactive sources is their flux as a function of distance. If the flux is emitted omnidirectionally from a point source, e.g., the sun, the flux density varies as the inverse square of the distance where

$$F_i = F_0 (4\pi r_0^2 / 4\pi r_i^2) = F_0 (r_0/r_i)^2$$

where

$4\pi r^2$ = area of sphere

F_i = flux at distance r_i , particles per unit time per unit area, $\text{m}^{-2} \text{s}^{-1}$

F_0 = flux at distance r_0 , particles per unit time per unit area, $\text{m}^{-2} \text{s}^{-1}$

r_0 = distance from source for flux F_0 , m

r_i = distance from source for flux F_i , m

9.4.2 Interaction Mechanisms

The three primary interactions of photons with matter are the photoelectric effect, Compton scattering, and pair production. *Photoelectric effect* is the interaction of an incident photon in which all of the energy is transferred to a bound electron, ejecting the electron from the atom. Thus, the kinetic energy of the resulting photoelectron is equal to the energy of the incident photon minus the binding energy of the electron. This mechanism is considered the principal transfer of energy by photons with energies below 50 keV and is less important at higher energies.

Compton scattering is the interaction of an incident photon in which it loses enough energy to an orbital or free electron to cause its ejection, with the remainder of the original photon's energy emitted as a new, lower energy photon. For a high-energy photon, there may be a series of Compton scattering events, each of which reduces the secondary photon energy before the sequence ends. This mechanism is considered the principal transfer of energy by photons with energies in the range of 50 keV to 5 MeV, i.e., gamma radiation.

Pair production is the interaction of an incident photon with the Coulomb force of the nucleus that produces an electron-positron pair. A positron is an unstable positively charged electron. Energy in excess of the equivalent rest mass of the two particles (1.02 MeV) appears as the kinetic energy of the pair and the recoil nucleus. The positron has a very short lifetime; it combines within 10^{-8} s with a free electron. The entire mass of these two particles is then converted to two gamma photons of 0.51 MeV energy each. Pair production is the dominant energy transfer mechanism for photons with energy greater than 5 MeV.

Figure 9.4 illustrates the relative importance of the three effects as a function of atomic number and energy.

9.4.3 Flux Density Attenuation

The attenuation of the flux density of a beam of photons can be represented by *scattering cross sections*, generally denoted by σ , that represent the probability of an interaction of the radiation with an atom in an absorber. The total scattering cross section for photons σ is given as the sum of the three principal effects by

$$\sigma = \sigma_{\text{photoelectric effect}} + \sigma_{\text{compton scattering}} + \sigma_{\text{pair production}} \quad (9.4)$$

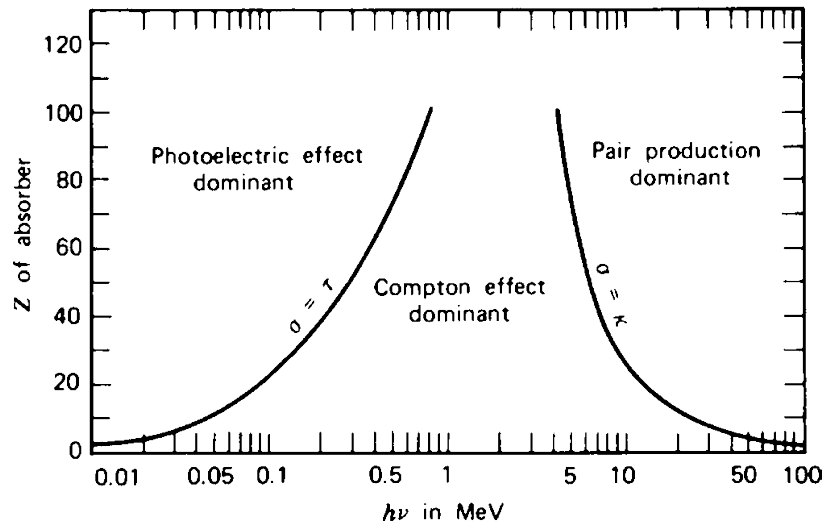


Fig. 9.4 Relative importance of the three photon interactions. [From Evans (1955); with permission of the McGraw-Hill Companies.]

where in typical units

σ = total scattering cross section of each atom in the absorber, $\text{cm}^2 \text{ nucleus}^{-1}$

$\sigma_{\text{photoelectric effect}}$ = photoelectric effect scattering cross section of each atom in the absorber, $\text{cm}^2 \text{ nucleus}^{-1}$

$\sigma_{\text{Compton scattering}}$ = Compton scattering cross section of each atom in the absorber, $\text{cm}^2 \text{ nucleus}^{-1}$

$\sigma_{\text{pair production}}$ = pair production scattering cross section of each atom in the absorber, $\text{cm}^2 \text{ nucleus}^{-1}$

If it is assumed that the particle interspacing is large so that all nuclei are exposed to the incoming radiation, as illustrated in Fig. 9.5, the collision rate or change in flux is

$$dI = -\sigma IN dx \quad (9.5)$$

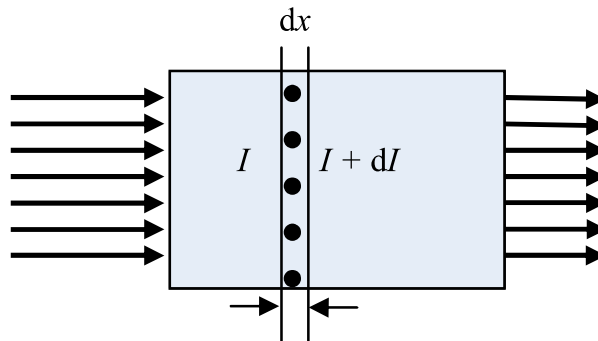


Fig. 9.5 Photon attenuation geometry.

where in typical units

dI = change in photon flux density, photons $\text{cm}^{-2} \text{s}^{-1}$

σ = scattering cross section, $\text{cm}^2 \text{nucleus}^{-1}$

I = photon flux density, photons $\text{cm}^{-2} \text{s}^{-1}$

N = number density of target nuclei, nuclei cm^{-3}

dx = infinitesimal target thickness, cm

Integration gives

$$\log_e I \Big|_{I_0}^I = -N\sigma x \Big|_0^x \quad (9.6)$$

so that

$$I = I_0 e^{-N\sigma x} = I_0 e^{-\mu x} = I_0 e^{-\mu_m \rho x} \quad (9.7)$$

where in typical units

I_0 = photon flux density at $x = 0$, photons $\text{cm}^{-2} \text{s}^{-1}$

$\mu \equiv N\sigma$ = linear attenuation coefficient, cm^{-1}

$\mu_m \equiv \mu \rho^{-1}$ = mass attenuation coefficient, $\text{cm}^2 \text{g}^{-1}$

ρ = density of absorbing material, g cm^{-3}

The attenuation is generally written in terms of either the *linear attenuation coefficient* μ or the *mass attenuation coefficient* μ_m . The latter is illustrated in Fig. 9.6 for a variety of elements and compounds. The attenuation coefficients represent the interaction of radiation with atoms in the absorber per unit length. Clearly, lead has a greater attenuation than aluminum. Absorption data for other materials, elemental and compounds, can be obtained from the National Institute of Standards and Technology (2007).

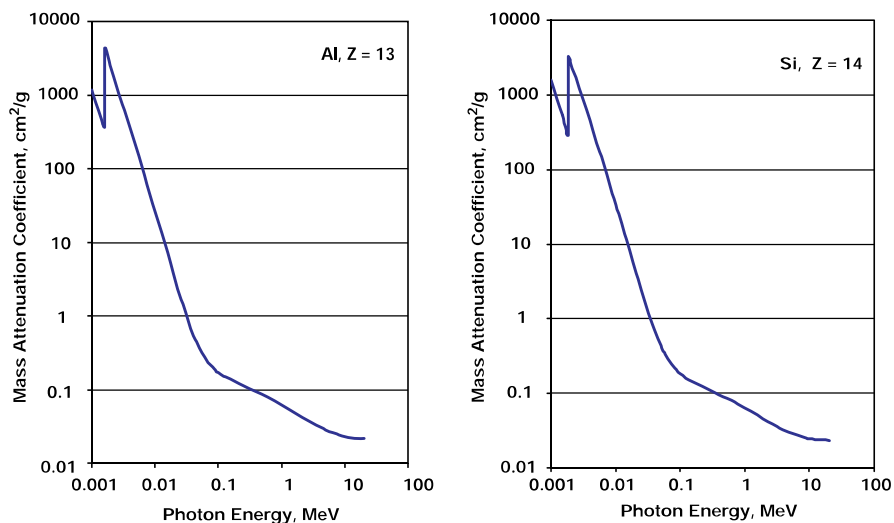


Fig. 9.6a Photon mass accommodation coefficients for aluminum and silicon. [Data from National Institute of Standards Technology (2007).]

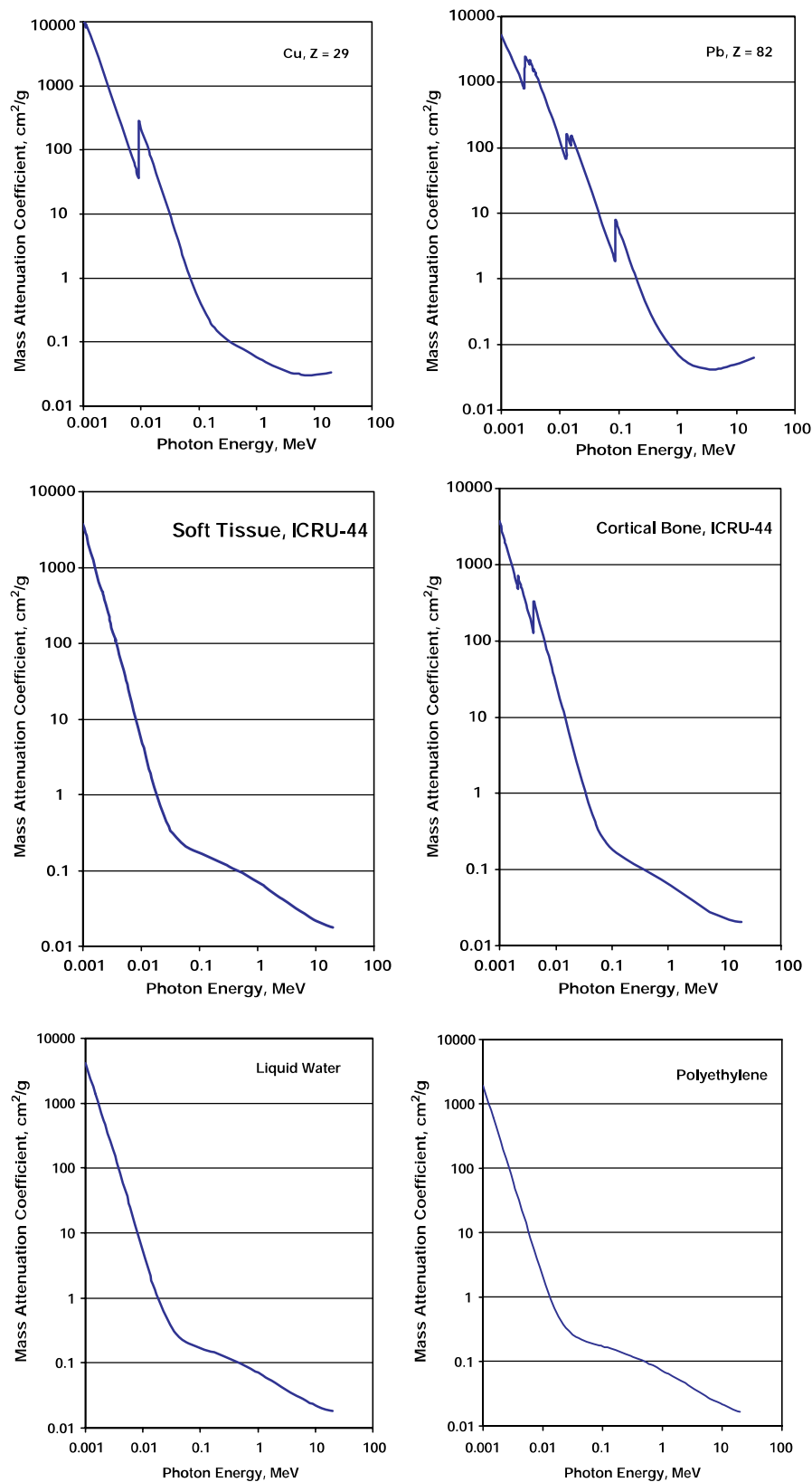


Fig. 9.6b Photon mass accommodation coefficients for copper, lead, soft tissue, cortical bone, liquid water, and polyethylene. [Data from National Institute of Standards Technology (2007).]

The attenuation for a graded material with the characteristics of each thickness x_i having linear attenuation coefficient μ_i , mass attenuation coefficient $\mu_{m,i}$ and density ρ_i is given by

$$I = I_0 \exp^{(-\mu_1 x_1 - \mu_2 x_2 - \mu_3 x_3 - \dots)} = I_0 \exp^{(-\sum \mu_i x_i)} = I_0 \exp^{(-\sum \mu_{m,i} \rho_i x_i)} \quad (9.8)$$

The attenuation for an alloy with fractional masses

$$X_i \equiv \frac{m_i}{\sum m_i}$$

follows as

$$I = I_0 \exp^{(-\mu_1 X_1 x - \mu_2 X_2 x - \mu_3 X_3 x - \dots)} = I_0 \exp^{(-\sum \mu_i X_i x)} = I_0 \exp^{(-\sum \mu_{m,i} \rho_i X_i x)} \quad (9.9)$$

Example 9.2

Problem: A 3-cm thick shield of aluminum attenuates the flux of a beam of photons by 40%. Determine the initial energy of the photons.

Solution: From Eq. (9.7) it follows that the mass attenuation coefficient is given by

$$\mu_m = -\frac{1}{x\rho} \log_e \left(\frac{I}{I_0} \right) = -\frac{\log_e 0.60}{3 \text{ cm} \times 2.7 \text{ g cm}^{-3}} = 0.0631 \text{ cm}^2 \text{ g}^{-1} \quad (9.10)$$

since the density of aluminum is 2.7 g cm^{-3} . This can then be used to find the energy of the photon in Fig. 9.6 to be $\approx 1 \text{ MeV}$.

The *half-value layer* (HVL) or *half-thickness value* (HTV) is defined as the distance a photon beam travels in an absorbing medium during which it loses half of its flux density. Substituting $I = I_0/2$ and $x = \text{HVL}$ into Eq. 9.7 gives

$$\text{HVL} = \frac{\log_e 2}{\mu} = \frac{\log_e 2}{\mu_m \rho} = \frac{0.69314}{\mu} = \frac{0.69314}{\mu_m \rho} \quad (9.11)$$

For each half-value layer that the photons traverse, the flux density is reduced by half. The photon flux density in terms of the half-value layer follows as

$$I = I_0 \left(\frac{1}{2} \right)^{x/\text{HVL}} \quad (9.12)$$

Figure 9.7 gives the half-value layer for several materials.

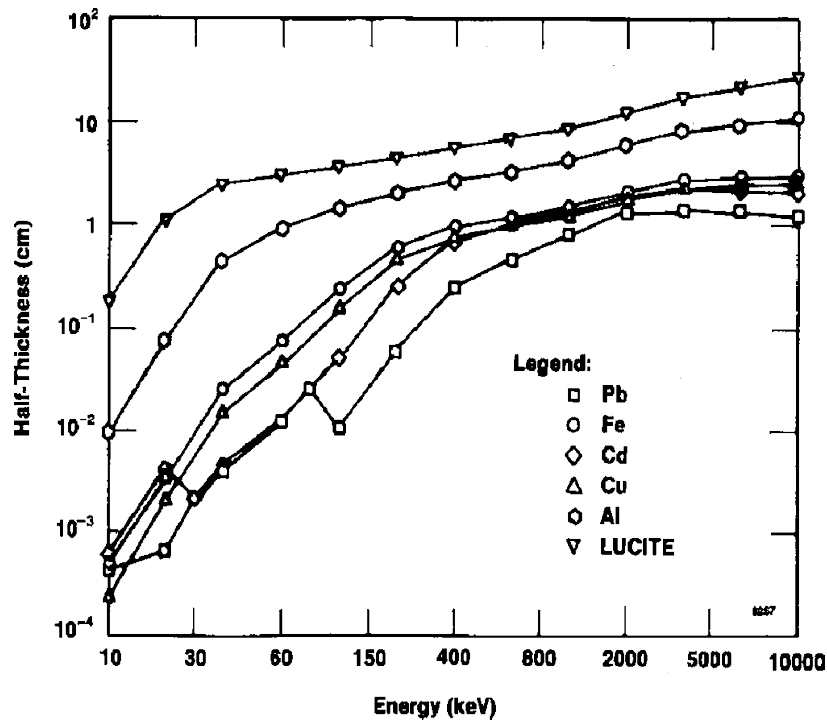


Fig. 9.7 Photon half-thickness values vs energy. [Courtesy of ORTEC (2007).]

9.5 Neutron Interactions

Neutrons are uncharged particles that have a mass slightly greater than a proton; the proton mass is $938.3 \text{ MeV } c^{-2}$ while the neutron mass is $939.6 \text{ MeV } c^{-2}$. While photons primarily interact with the electron cloud surrounding the nucleus of the atom of the material in which they are traveling, neutrons primarily interact directly with the nucleus, and so their attenuation is different from photons. Photon attenuation is dictated primarily by the density of material, so that dense materials such as lead are good photon absorbers. Because the nucleus is on the order of 10^{-14} m and the atom is on the order of 10^{-10} m , the probability of an interaction of a neutron with the nucleus is small, so that neutrons can travel large distances in a medium before an interaction occurs.

Neutrons interact with a material primarily by elastic scattering, inelastic scattering, and neutron capture. In *elastic scattering* (classical Rutherford scattering), energy is conserved in the interaction where energy lost by the neutron is transferred to the nucleus, which is then called a *recoil nucleus*. This is the most important interaction for neutron energies in the range of 1–10 MeV. Maximum energy is transferred if the mass of the nucleus is equal to the mass of the neutron. Consequently, hydrogen with an atomic number of one (nucleus of one proton) is a good absorber of neutrons. Lead with an atomic number of 82 (82 protons and 126 neutrons) is not. Thus, materials with high hydrogen content, such as plastics or water, are effective in slowing fast neutrons. In *inelastic scattering*, the energy of the neutrons is absorbed by the target nuclei that then emit photons. The probability of this occurring increases with the size of the nucleus and energy of the neutron. *Neutron capture* occurs when the neutrons have lost sufficient energy

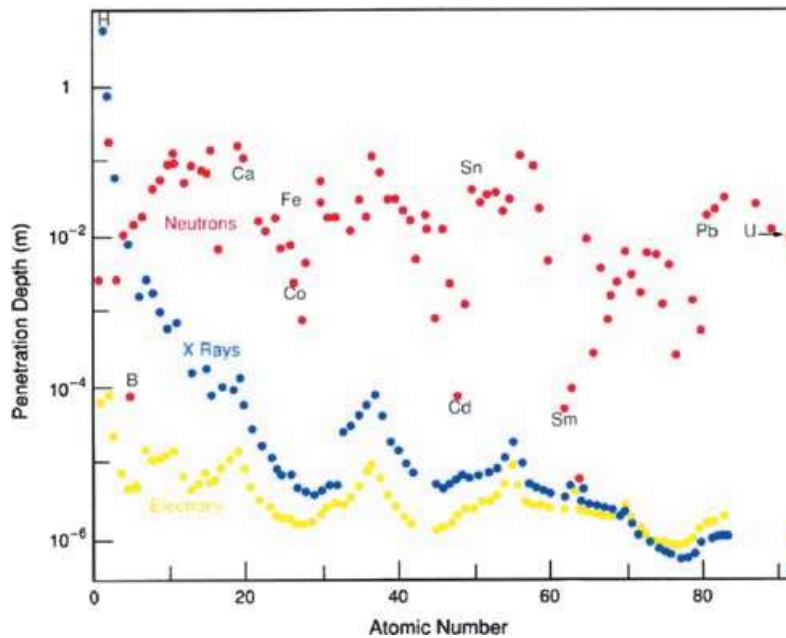


Fig. 9.8 Penetration depth for low-energy electrons, x-rays, photons, and thermal neutrons. [From Pynn (2007); courtesy of the Los Alamos National Laboratory.]

and are captured by nuclei forming an isotope with the atomic mass number increased by one. This isotope is usually unstable and, if so, then emits a particle or photon or results in fission of the nucleus. The probability of capture is inversely proportional to the energy of the neutron. The total microscopic cross section is the sum of the cross sections for each of the different processes (elastic scattering, inelastic scattering, and neutron capture) and is a function of the energy of the neutron and the atomic number of the nuclei.

Neutron shielding calculations are more complicated than those for photons and are generally carried out using sophisticated Monte Carlo radiation transport codes. However, shielding calculations for low flux density neutron sources can be performed using the exponential relationships developed for photons if the attenuation factor is available for the specific energy of the neutron interacting with the specific absorbing material. Figure 9.8 gives the penetration depth for low-energy electrons, x-rays, and thermal neutrons. While there is a consistency for penetration depth of the electrons and photons, there is no systematic dependence on atomic number. The linear attenuation coefficient for thermal neutrons can be estimated from Fig. 9.8.

9.6 Charged Particle Attenuation

9.6.1 Interaction Mechanisms

Charged particles traveling in a medium interact with the electrons of an atom by the electrostatic (Coulomb) force by several mechanisms, including excitation, ionization, and Bremsstrahlung. In *excitation*, the charged particle transfers some of its energy to the electrons with insufficient energy to ionize them. When the electrons decay, they emit photons called fluorescence. In *ionization*, energy

transferred to the electrons is sufficient to cause ionization, and so electrons are removed from the atom forming electron-ion pairs. The *mean ionization energy* or the *mean ionization potential* is the minimum energy required to form electron-ion pairs. If the electrons created by the primary interactions produce additional ionizations, these are called *delta rays*. *Bremsstrahlung* (German for braking radiation) is the name given to the electromagnetic radiation emitted as photons when a fast-moving charged particle loses energy upon being accelerated and deflected by the electric field surrounding a positively charged atomic nucleus. This is the mechanism by which x-rays are produced in an x-ray machine.

9.6.2 Stopping Power and Linear Energy Transfer

Stopping power, S , is defined as the average energy lost by a charged particle per unit distance traveled, generally given in $\text{keV } \mu\text{m}^{-1}$. *Linear energy transfer*, LET , is the average energy locally deposited in an absorber per unit path length, generally given in $\text{keV } \mu\text{m}^{-1}$ or $\text{keV mg}^{-1} \text{cm}^2$. The concepts of stopping power and LET are closely related; the difference is that stopping power refers to the change in energy of the radiation, while LET refers to the energy deposited in the absorber. While stopping power and linear energy transfer have the same units, they may not be equal because not all of the energy removed from the radiation may be deposited locally and may not contribute to LET so that $LET \leq S$. For example, Bremsstrahlung or excitation may produce photons that can leave the local volume. In practice, the difference between LET and S is usually small so that a good approximation is that they are equal except possibly for microscopic volumes. It follows then that

$$S \equiv -\frac{dE}{dx} \quad (9.13)$$

$$LET \equiv -\frac{dE_d}{dx} \approx -\frac{dE}{dx} \quad (9.14)$$

$$LET_\rho \equiv -\frac{1}{\rho} \frac{dE_d}{dx} \approx -\frac{1}{\rho} \frac{dE}{dx} \quad (9.15)$$

where in typical units

S = stopping power, average energy lost per path length, $\text{keV } \mu\text{m}^{-1}$

LET = average energy absorbed per path length, $\text{keV } \mu\text{m}^{-1}$

LET_ρ = average energy absorbed per path length per density, $\text{keV mg}^{-1} \text{cm}^2$

E = energy of the charged particle, keV

E_d = energy deposited in absorber, keV

x = linear distance, cm

ρ = density of absorber, g cm^{-3}

Quantitative assessments of stopping power are generally obtained by complex Monte Carlo radiation transport codes. However, there are analytical quantitative assessments specific to particular energy ranges, types of particles, and absorbing

materials. As an example, the Bethe–Bloch formula for the stopping power of heavy charged particles, such as the alpha particle and proton, but not electron, is

$$-\frac{dE}{dx} = \frac{4\pi k_e^2 Z^2 e^4 n}{m_e c^2 \beta^2} \left[\log_e \left(\frac{2m_e c^2 \beta^2}{I(1-\beta^2)} \right) - \beta^2 \right] \quad (9.16)$$

where

A = atomic mass number of absorber

c = speed of light in vacuum, m s^{-1}

dE/dx = stopping power, J m^{-1}

e = magnitude of charge of an electron, C

I = mean excitation energy of the absorber, J

k_e = Coulomb's constant, $\text{N m}^2 \text{C}^{-2}$

m_e = electron rest mass, kg

$n = N_A \rho z/A$, number of absorber electrons per unit volume, electrons m^{-3}

N_A = Avogadro's number, kg kmol^{-1}

v = velocity, m s^{-1}

Z = atomic number of the incident particle

$\beta \equiv v c^{-1}$

ρ = density of absorber, kg m^{-3}

The excitation energy is the addition of a discrete amount of energy to a system, such as an atomic nucleus, an atom, or a molecule, that results in its alteration, ordinarily from the condition of lowest energy (ground state) to one of higher energy (excited state). The mean excitation energies of an absorber material can be approximated in terms of the material's atomic number Z by

$$\begin{aligned} I &\approx 19.0 \text{ eV}; Z = 1 \text{ (hydrogen)} \\ I &\approx 11.2 \text{ eV} + (11.7)(Z) \text{ eV}; 2 \leq Z \leq 13 \\ I &\approx 52.8 \text{ eV} + (8.71)(Z) \text{ eV}; Z > 13 \end{aligned} \quad (9.17)$$

or by

$$I = 16 \cdot Z^{0.9} \text{ eV} \quad (9.18)$$

From the Bethe–Bloch equation, it can be seen that the specific energy loss is greatest for high-density, high- Z absorbers and also for ions in higher charge states. For low energies, i.e., for small velocities of the particle so that $\beta \ll 1$, the energy loss decreases as v^{-2} and reaches a minimum for approximately $E = 3 \text{ mc}^2$. For high-velocity relativistic particles where $\beta \rightarrow 1$, the energy loss increases logarithmically.

For alloys or graded materials, weighted contributions should be used based on

$$\log_e I = \frac{1}{n} \sum_i n_i Z_i \log_e I_i \quad (9.19)$$

where in typical units

I = mean excitation energy of the alloy or graded material, eV

n = total number of electrons per unit volume of alloy or graded material
electrons m^{-3}

I_i = mean excitation energy of the material i , eV

n_i = electron density of material i , electrons m^{-3}

Z_i = atomic number of material i

Example 9.3

Problem: Determine the mean excitation energy of water, H_2O .

Solution: The atomic number of hydrogen is $Z_H = 1$ and oxygen $Z_O = 8$, so that Eq. (9.17) for hydrogen and oxygen gives for their excitation energies

$$I_H = 19.0 \text{ eV} \quad (9.20)$$

$$I_O = 11.2 + 11.7 \times 8 = 104.8 \text{ eV} \quad (9.21)$$

Then Eq. (9.19) gives for the mean excitation energy of water

$$\begin{aligned} I &= \exp \left(\frac{1}{n} \sum_i n_i Z_i \log_e I_i \right) \\ &= \exp \left(\frac{2 \times 1}{10} \log_e 19.0 + \frac{1 \times 8}{10} \log_e 104.8 \right) = 74.5 \text{ eV} \end{aligned} \quad (9.22)$$

Equation (9.16) can be rewritten in a more computationally friendly form as

$$-\frac{dE}{dx} = \frac{5.08 \times 10^{-32} Z^2 n}{\beta^2} \left[\log_e \left(\frac{1.02 \times 10^6 \beta^2}{I(1 - \beta^2)} \right) - \beta^2 \right] \quad (9.23)$$

The important parameters are velocity v , atomic number of incident particle Z , number of electrons per unit volume, and the mean excitation energy of the absorber in which the particle is traveling I .

Example 9.4

Problem: Determine the stopping power of water for protons with kinetic energy of 1 MeV.

Solution: The number density of electrons in water is given by

$$\begin{aligned} n_{\text{H}_2\text{O}} &= \frac{N_A \rho_{\text{H}_2\text{O}} \eta_{\text{H}_2\text{O}}}{M_{\text{H}_2\text{O}}} = \frac{(6.02 \times 10^{26}) \times 10^3 \times 10}{18.0} \\ &= 3.344 \times 10^{29} \text{ electrons/m}^3 \end{aligned} \quad (9.24)$$

where

$n_{\text{H}_2\text{O}}$ = number density of electrons, electrons m^{-3}

N_A = Avogadro's number, molecules kmol^{-1}

$\rho_{\text{H}_2\text{O}}$ = density of water = 1000 kg m^{-3}

$\eta_{\text{H}_2\text{O}}$ = number of electrons per molecule of water, 10 electrons molecule $^{-1}$

$M_{\text{H}_2\text{O}}$ = molecular mass, 18 kg kmol^{-1}

The velocity ratio β can be determined from the nonrelativistic kinetic energy E_k and the rest energy of the proton by

$$E_k = \frac{1}{2} m_0 v^2 = \frac{1}{2} m_0 c^2 \beta^2 = \frac{1}{2} E_0 \beta^2 \quad (9.25)$$

$$\beta = \left(\frac{2E_k}{E_0} \right)^{1/2} = \left(\frac{2 \times 1}{938.3} \right)^{1/2} = 0.04617 \quad (9.26)$$

where

E_k = kinetic energy of the proton, 1 MeV

$E_0 = m_0 c^2$, rest energy of the proton, 938.3 MeV

m_0 = rest mass, MeV c^{-2}

For a proton $Z = 1$, so that Eq. (9.23) gives

$$\begin{aligned} -\frac{dE}{dx} &= \frac{5.08 \times 10^{-32} (1)^4 (1^2) 3.344 \times 10^{29}}{(0.04617)^2} \\ &\times \left[\log_e \left(\frac{1.02 \times 10^6 (0.04617)^2}{74.5 (1 - (0.04617)^2)} \right) - (0.04617)^2 \right] \\ &= 27 \text{ keV } \mu\text{m}^{-1} \end{aligned} \quad (9.27)$$

Bethe also developed a formula, not given here, to determine the stopping power of low-mass charged particles such as electrons. Figure 9.9 illustrates the magnitudes of the various contributions to the stopping power of protons and electrons in aluminum and stopping power of a variety of elementary particles in water.

Figure 9.10 illustrates the LET spectra of the galactic cosmic rays behind aluminum shielding of 5 g cm^{-2} at a modest level of solar activity of $F_{10.7} = 150$

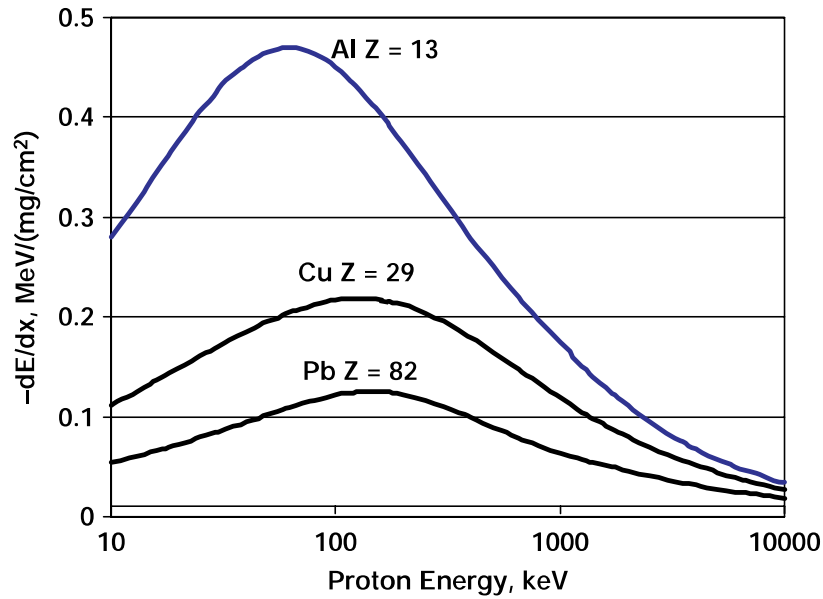


Fig. 9.9 Stopping power of protons in aluminum, copper, and lead in water.

in polar orbit illustrating the effect of the geomagnetic field. The galactic cosmic ray mode used is CRÈME-86 (Space Environment Information System 2007).

The range R , or depth of penetration, of a charged particle in an absorbing material can be expressed in terms of LET by

$$R = \int_0^x dx = \int_E^0 \left(\frac{dx}{dE} \right) dE = - \int_0^E \left(\frac{dx}{dE} \right) dE = \int_0^E \frac{dE}{S} \quad (9.28)$$

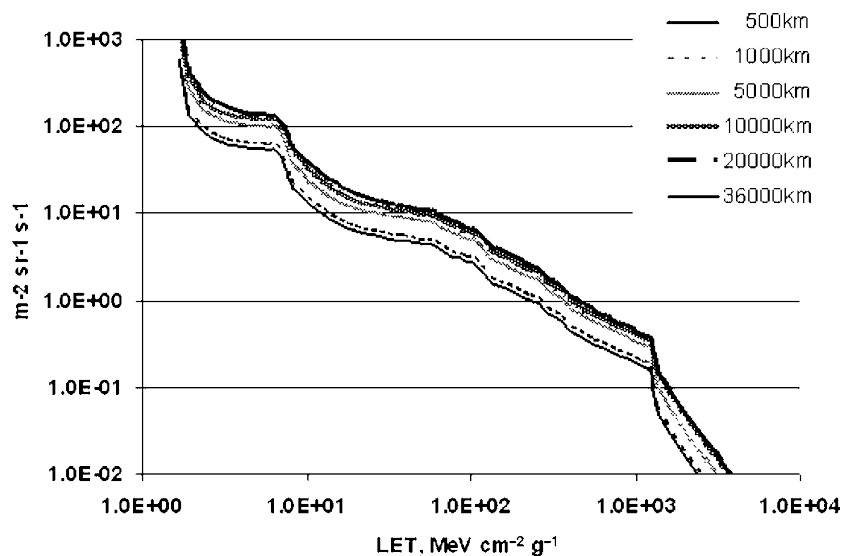


Fig. 9.10 Galactic cosmic ray LET spectrum, as a function of altitude for shielding of 5 g cm^{-2} , $F_{10.7} = 150$, and polar orbit.

$$= \int_0^E \frac{dE}{S} \approx \int_0^E \frac{dE}{LET} = \frac{1}{\rho} \int_0^E \frac{dE}{LET_\rho} \quad (9.29)$$

where in typical units

R = range or depth of penetration, cm

ρ = density of material, g cm⁻³

$LET_\rho \equiv \rho^{-1} LET$, eV g⁻¹ cm²

If the energy lost or absorbed is independent of the energy of the particle, then the equation can be integrated treating the integrand as a constant, so that

$$R = \frac{E}{S} \approx \frac{E}{LET} = \frac{E}{\rho LET_\rho} \quad (9.30)$$

Figures 9.11 and 9.12 illustrate the average penetration depth of protons and electrons in aluminum using the SRIM radiation transport code (Ziegler 2007) for protons and the Casino radiation transport code (Drouin 2007) for electrons.

An advantage of charged particle therapy in tumor treatment is illustrated in Figure 9.13. It illustrates that x-ray (photon) radiation loses energy rapidly as it penetrates tissue. Conversely, charged particles, such as ion or proton radiation, deposit most of their energy at a specific location as the radiation slows down to an energy called the Bragg peak. This means that high-radiation doses can be delivered at a tumor site, sparing surrounding healthy tissue.

9.7 Radiation Effects on Materials

Energetic radiation can penetrate analog and digital components, causing temporary faults or permanent failures. The effects of radiation on electronic devices

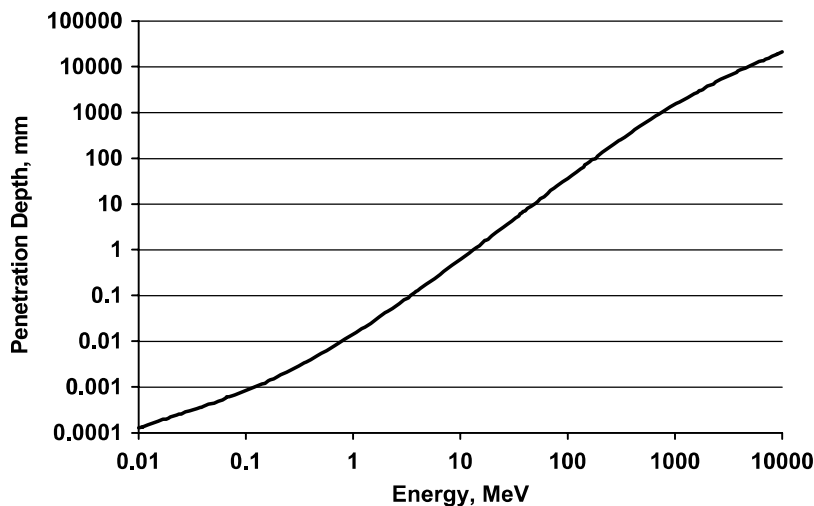


Fig. 9.11 Average proton penetration depth in aluminum.

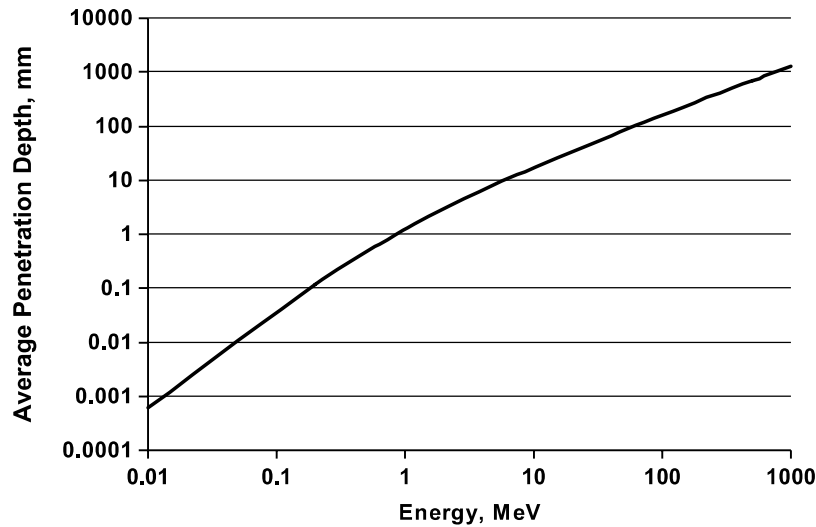


Fig. 9.12 Average electron penetration depth in aluminum.

can be classified into three primary types: displacement damage, total ionizing dose, and single-event effects. Each is discussed in turn.

9.7.1 Displacement Damage

Displacement damage (DD) is the result of radiation that causes *atomic displacements* when radiation interacts with atomic nuclei, displacing or removing them from their lattice sites. This upsets the periodicity of the lattice in the material, creating lattice defects, as illustrated in Fig. 9.14. Displacement damage, also known as *bulk damage*, is caused by the cumulative effects of non-ionizing radiation that include protons and ions at all energies, electrons greater than about 150 keV, and neutrons from onboard radioactive power sources or

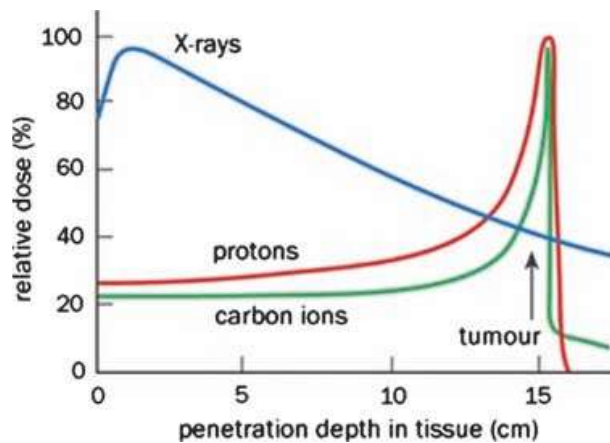


Fig. 9.13 Advantage of charged particle therapy. [From Institute of Physics (2003); courtesy of *Physics World*.]

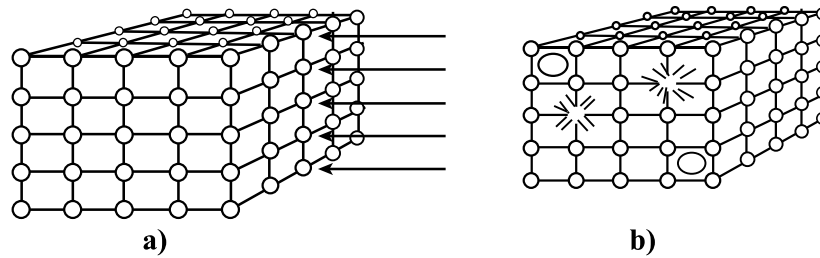


Fig. 9.14 Displacement damage: a) Atoms in a crystal are arranged in an orderly manner before bombardment by electrons. b) Incoming electrons displace atoms and cause the crystal to have poor electrical properties.

secondary particles from the initial interactions. The most important consequence of displacement damage in a semiconductor is a reduction in the lifetime of the minority carrier. In a p-type semiconductor, the minority carrier is the electron and in an n-type semiconductor the minority carrier is the hole. Because displacement damage is caused by a broad spectrum of energies, it is convenient to select a standard particle energy and assign to this a damage effectiveness of unity. Consequently, particles at other energies have a higher or lower relative damage effectiveness factor. A common standard is 1-MeV electrons, where the functional damage due to a particle spectrum can be expressed in simple terms, using the unit *equivalent 1-MeV electrons per cm²*. Solar cells are especially susceptible to displacement damage with potentially significant losses to their power output, as illustrated in Fig. 9.15, which illustrates the reduction in maximum power output.

Figure 9.16 illustrates displacement damage in a voltage regulator.

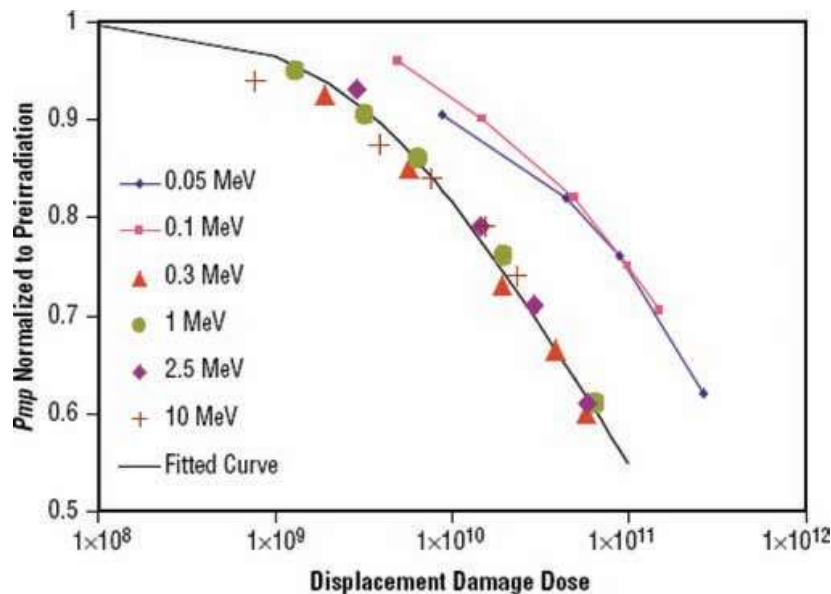


Fig. 9.15 Effect of displacement damage on multijunction solar cell maximum power under proton irradiation. [From Walters et al. (2004); courtesy of NASA.]

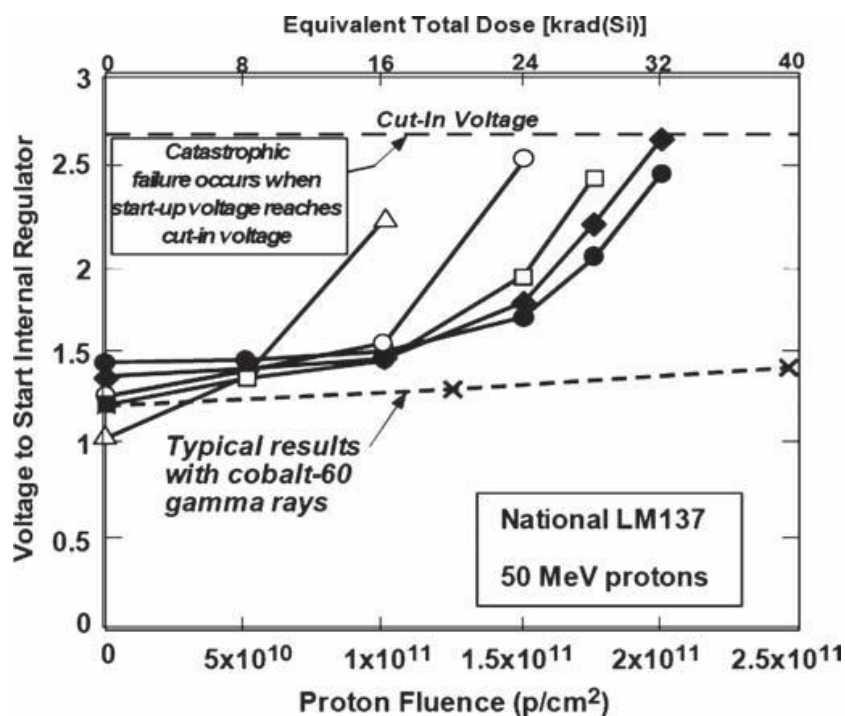


Fig. 9.16 Displacement damage in a voltage regulator. [From Johnson (2007); courtesy of NASA, Jet Propulsion Laboratory.]

9.7.2 Total Ionization Dose

Total ionization dose (TID) is the accumulated effect of ionizing radiation over the duration of a space mission. Ionization radiation generates electron-hole pairs in semiconductors and in insulators such as silicon dioxide. It is possible for the electrons and holes to recombine or be transported away by an electric field. However, holes that have lower mobility than electrons are often trapped at interfaces between the semiconductor and insulator or within the bulk material. This can cause an increase in the leakage currents that can increase power consumption, changes to the time constant of the device, a reduction in the gain, and changes to the threshold voltage of a metal-oxide semiconductor (MOS) gate. With increasing exposure, there is a continuing decrease in functionality until the device eventually fails.

Figure 9.17 of an NMOS transistor illustrates the gate oxide and conducting channel formed between the source and drain. Trapped positive charges shown in the gate oxide cause a shift in the threshold voltage that can ultimately lead to increases in the electric field, causing current leakage. Similar to the effect on the gate oxide, the field oxide can trap negative charges. If sufficient charge is trapped, a conducting channel of electrons will form, connecting source to drain and effectively shorting the two formerly isolated device elements, causing a failure.

Total ionization dose (TID) is measured by the *absorbed dose*, which is the ionization energy absorbed by matter per unit mass. The unit utilized is preferably the gray (Gy), which is defined as the deposition of one joule of energy per kilogram of mass. Also used is the rad (radiation absorbed dose), where 1 Gy = 100 rad. Commercial digital devices can normally tolerate a few Krad without

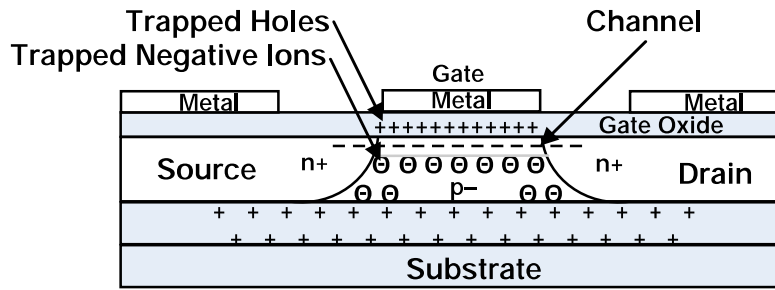


Fig. 9.17 NMOS transistor with trapped charges.

significant effect. Modern submicron technologies tend to be more resistant to total ionization dose effects than older technologies. Satellites and space probes typically begin to encounter TID effects at about 2 krad (Si) for commercial parts and about 20 krad (Si) for semihard parts. Below 500 km, commercial parts used in many student satellite programs do perform adequately. It should be noted that the total ionization dose increases substantially for higher inclination orbits. Figure 9.18 illustrates the annual absorbed dose in silicon in a spherical shell of given thickness for a polar orbit as a function of altitude.

9.7.3 Single-Event Effects

Single-event effects (SEE) are transient or permanent effects that result from a single or a few quanta of radiation. Their effects on electronic devices are normally divided into four categories characterized by the performance defect.

Single-event upset (SEU) is a radiation-induced transient or soft error in electronic devices caused by the ionization trail of electron-hole pairs that results from the transit of a quantum of radiation. It can manifest itself as a transient pulse in an analog device or circuit or as a bit flip in a digital logic device. These are soft errors in that the device will otherwise continue to perform normally. Error

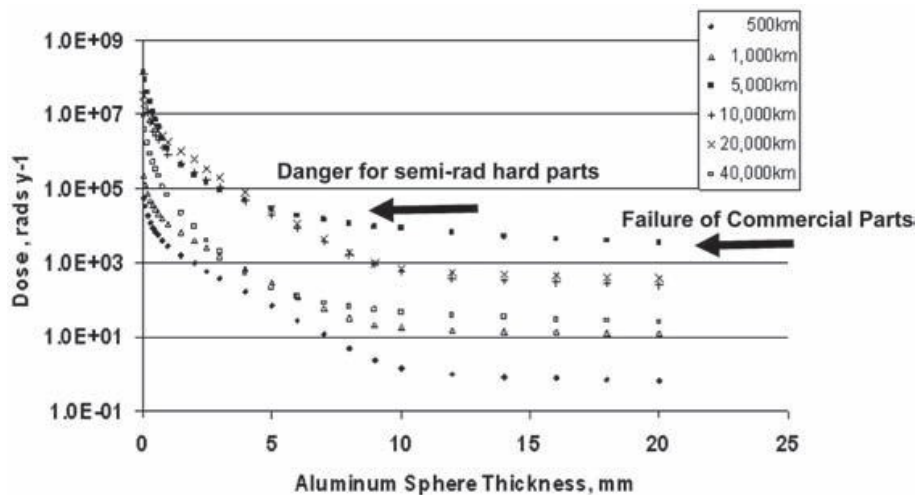


Fig. 9.18 Absorbed dose for silicon in a spherical shell as function of altitude for polar orbit.

detection and correction codes are often utilized to reduce the effects of SEUs in digital logic devices, especially semiconductor memories.

Single-event hard error (SEHE) is a radiation-induced permanent change in the operation of electronic devices caused by the ionization trail of electron-hole pairs that results from the transit of a quantum of radiation. An example would be a permanently stuck bit in a logic device.

Single-event latchup (SEL) is a radiation-induced condition that results in excessive power draw in electronic devices caused by the ionization trail of electron-hole pairs from the transit of a quantum of radiation. The ionization trail causes a short circuit between the power supply and ground in semiconductor devices that can result in large currents if not externally limited. A shutdown and repowering of the circuit is generally sufficient to correct the problem, if the power was limited. If power is not limited or quickly removed, catastrophic failure may occur due to excessive heating and failure of the metallization or wire bonds.

Single-event burnout (SEB) is a radiation-induced condition that results in excessive power draw in a power transistor caused by the ionization trail of electron-hole pairs from the transit of a quantum of radiation. This is a permanent destructive failure. SEBs include burnouts of power metal oxide semiconductor field effect transistors (MOSFETs) and gate ruptures.

The deposition of electron-hole pairs in a semiconductor is illustrated in Fig. 9.19, creating a conducting channel between the two layers separated by the depletion region.

The occurrence of single-event effects have become more likely as devices are reduced in size with increased numbers of diodes and transistors so that the critical charge required to cause an event decreases. Table 9.1 ranks the susceptibility of current technologies to SEEs. Table 9.2 identifies several techniques that can be employed to mitigate single event errors.

As discussed in Chapter 1, derating can extend part life and increase overall system reliability. Some organizations have developed guidelines for derating radiation limits on parts based on the age, part category, test method, changes in the design or manufacturing process, variability within a lot, variability from lot to lot, and other factors. The radiation hardness of a part from one manufacturer is not generally the same as from another. Even though a device may have the same name, part number, and electrical characteristics, the designs and processes of the part may be different, and so their radiation hardness characteristics may be

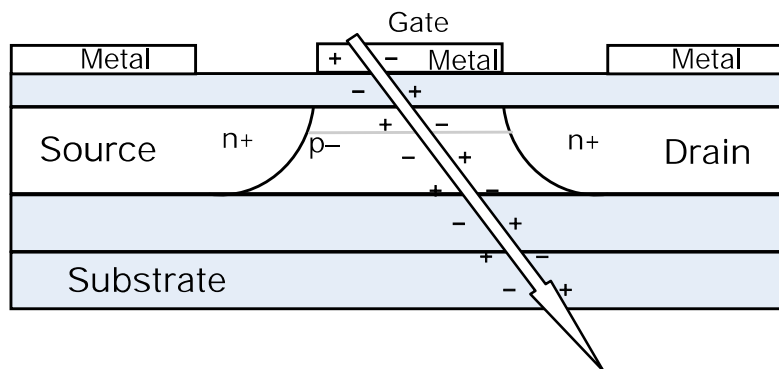


Fig. 9.19 Deposition of energy in a semiconductor device.

Table 9.1 Susceptibility of semiconductor technologies to SEE

Susceptibility	Technology
Least susceptible	Radiation hardened CMOS/SOS or CMOS/SOI (complementary metal oxide semiconductor/silicon-on-sapphire or silicon-on-insulator) CMOS/SOS or CMOS/SOI CMOS Standard bipolar Low power Schottky bipolar
Most susceptible	NMOS DRAMs (N-channel metal–oxide semiconductor dynamic random access memories)

different, which is especially true for different manufacturers. While the design of a part that has been available for some time and flight qualified may have not changed, often the manufacturing process may have changed that could have significant consequences on its radiation hardness. In general, digital parts and integrated circuits tend to evolve very rapidly, making frequent testing more necessary. Analog parts tend to evolve more slowly.

Figure 9.20 shows the spatial distribution of single-event upsets in the UoSAT-2 (AMSAT-Oscar-11) satellite launched in March 1984 from Vandenberg Air Force Base, California. The orbit had a perigee altitude of 679 km, apogee altitude of 697 km, and an inclination of 98.7 deg. Note the increased occurrences of single-event upsets from protons in the South Atlantic Anomaly and galactic cosmic rays in the polar regions. The upsets are measured in a dynamic random access memory (DRAM).

The Air Force Space Test Program's Advanced Photovoltaic and Electronic Experiments (APEX) Mission was launched in 1994. The APEX spacecraft has a perigee altitude of 361 km, an apogee altitude of 1952 km, and an orbital inclination of 70 deg. A secondary payload was the Cosmic Ray Upset Experiment (CRUX) developed by the Goddard Space Flight Center (GSFC) in Maryland to make observations to validate and update modeling techniques that predict upset rates in microcircuit memory devices (Mullen 1995). There is a total of 80 Mbits of data storage in 164 devices in CRUX. The upsets illustrated in Fig. 9.21 are

Table 9.2 Mitigation strategies for single-event effects

Protection technique	Procedure
Error detecting and correction codes	Error detection and correction codes can be employed to confirm stored data.
Watchdog timer	If the timer is not reset in a predetermined period of time, the system is rebooted from a read-only memory.
Redundancy	If two systems do not give the same results, both systems are rebooted from a read-only memory.
Majority voting	If three or more systems do not give the same results, the majority result is accepted and the disagreeing systems or systems are rebooted from a read-only memory.

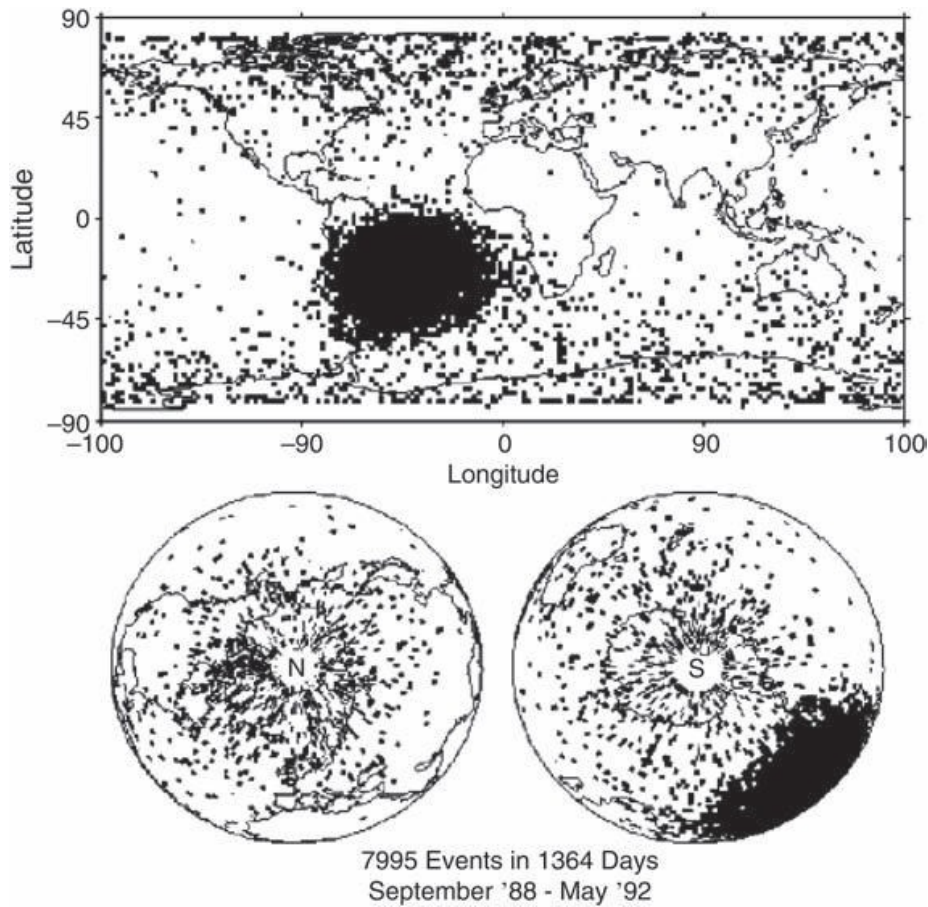


Fig. 9.20 Single-event upsets in the UoSAT-2 spacecraft. [From Dyer (2002); courtesy of the European Space Agency.]

those observed in 16 Hitachi 1 Mbit static random access memories (SRAM). The effect of increased altitude can be clearly seen, but the orbital inclination limited the coverage to geographic latitudes of ± 70 deg.

9.7.4 Charge Deposition

The assessment of the potential susceptibility of an electronic part to radiation damage can be determined by estimating the amount of charge that the radiation deposits. The charge deposition for a radiation with given LET for a material with distance traveled, s , is

$$E_{\text{deposited}} = LET_{\rho} \times s \times \rho \quad (9.31)$$

where for typical units

- $E_{\text{deposited}}$ = energy deposited, eV
- LET_{ρ} = linear energy transfer, $\text{eV g}^{-1} \text{cm}^2$
- s = distance traveled, cm
- ρ = density, g cm^{-3}

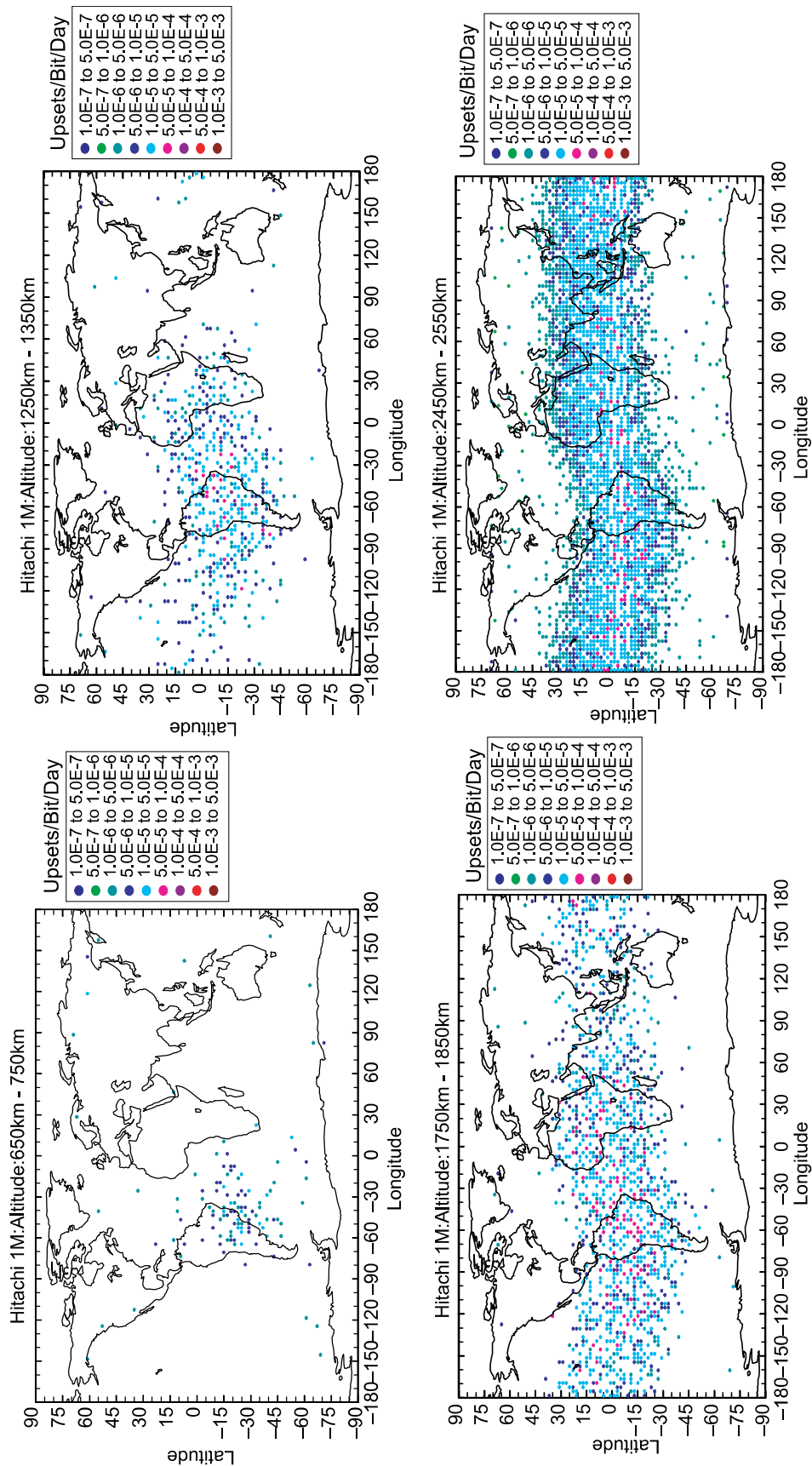


Fig. 9.21 SRAM upset rates on CRUX/APEX. [From Adolphsen et al. (1995); copyright 1995 IEEE.]

The charge deposited is then

$$Q = \frac{E_{\text{deposited}} e}{I_1} = \frac{LET_\rho \times s \times \rho \times e}{I_1} \quad (9.32)$$

where

Q = charge deposited, C

e = magnitude of electron charge, C

I_1 = first ionization energy of the atoms in the absorber, eV

The first ionization energy or potential is the energy required to remove one or more electrons from a neutral atom to form a positively charged ion. Table 9.3 gives the energy required to create an electron-hole pair for a variety of semiconductors and insulators used in electrical devices.

Example 9.5

Problem: Determine the charge deposited per micron in silicon by a particle with an LET of $10 \text{ MeV cm}^2 \text{ mg}^{-1}$.

Solution: It follows directly from Table 9.3 that for silicon $I_1 = 3.62 \text{ eV}$ and the density of silicon is 2.329 g cm^{-3} , so that Eq. (9.32) gives

$$\begin{aligned} \frac{Q}{s} &= \frac{LET_\rho \times e \times \rho}{I_1} \\ &= \frac{(1.0 \times 10^{10} \text{ eV cm}^2 \text{ g}^{-1})(1.6 \times 10^{-19} \text{ C})(2.329 \text{ g cm}^{-3})}{3.62 \text{ eV}} \\ &= 1.0 \times 10^{-9} \text{ C cm}^{-1} = 0.10 \text{ pC } \mu\text{m}^{-1} \end{aligned} \quad (9.33)$$

Table 9.3 Properties of semiconductors and insulators

Material	Chemical symbol	Type	Atomic/ molecular mass	Density, g cm^{-3}	First ionization energy, eV
Silicon	Si	Semiconductor	28.0855	2.329	3.62
Germanium	Ge	Semiconductor	72.640	5.323	2.98
Gallium arsenide	GaAs	Semiconductor	144.645	5.776	4.8
Cadmium telluride	CdTe	Semiconductor	240.011	6.200	4.43
Silicon dioxide	SiO_2	Insulator	60.084	2.4	17.0
Silicon nitride	Si_3N_4	Insulator	140.283	3.29	10.8
Aluminum oxide	Al_2O_3	Insulator	101.961	3.69	19.1

9.7.5 Radiation Hardness Assurance and Qualification

Assurance of reliable continuous operations of electronic systems in a radiation environment is complicated, time consuming, and expensive. Commercial-off-the-shelf (COTS) electronics components are generally not qualified for radiation environments above an absorbed dose of a few kilorads. Consequently, components must often be designed, qualified, and/or shielded for the anticipated radiation environment in which they will be employed. Because of a limited market, there are few sources of radiation hard components, and when available, they are expensive, requiring long delivery times. As a result, there is an increasing reliance on COTS parts that are either derated or shielded so that they can perform in an intensive radiation environment. The effective use of COTS parts requires that an appropriate formal radiation hardness assurance and testing program be employed if risk is to be adequately addressed.

Table 9.4 illustrates the typical radiation issues for selected electronic part types.

Table 9.5 gives some typical radiation test criteria, primarily to qualify commercial electronic parts. Often a proton or neutron beam test is adequate to simultaneously evaluate all three criteria.

Radiation hardness assurance addresses the management of the risk from radiation of the system under consideration by an active discipline initiated at the feasibility phase. If initiated later in the project, the impact can be invasive and costly. A top-level summary of the basic steps of a radiation hardness assurance program is given in Table 9.6. Design margins are employed because of uncertainty of the environmental assessment, pedigree of the part, expected part-to-part

Table 9.4 Sample device types and sensitive areas

Device type ^a	Sensitive area	Single-event effect
Memories	Memory cells Control logic	Bit flips Bit flips if sequential, transients if combinatorial
Combinatorial logic	Combinatorial logic	Transients
Sequential logic	Sequential logic	Bit flips
FPGAs	Combinatorial logic Sequential logic	Transients Bit flips
Microprocessors	Registers, cache, sequential control logic Combinatorial control logic	Bit flips Transients
ADCs, DACs	Analog portion Digital portion	Transients Bit flips or transients depending on design
Linear ICs	Analog area	Transients
Photodiodes	Photodiode	Transients

^aFPGA = Field programmable gate array, ADC = analog-to-digital converter, DAC = digital-to-analog converter, and IC = integrated circuit.

Source: From LaBel (1996); courtesy of NASA.

Table 9.5 Radiation test criteria

Radiation categories	Particle types	Radiation tests
Total ionization dose	Photons	Total dose, gamma, grays
Non-ionization energy loss (NIEL)	Neutrons	Total fluence, 1-MeV equivalent neutrons cm ⁻²
Single-event effects	Heavy ions	Total fluence ≥20-MeV heavy ions cm ⁻²
Joint test	Protons, neutrons	High-energy protons or neutrons ≥ dose and fluence

Table 9.6 Basic steps in radiation hardness assurance program

Step	Description
1	Identify the radiation exposure
2	Identify acceptable risk
3	Determine margins of safety
4	Identify potentially susceptible parts, maintain parts list
5	Procure rad-hard parts if possible
6	Identify parts at risk
7	Establish risk mitigation strategies for parts at risk
8	Confirm hardness of rad-hard parts by analysis and test
9	Confirm risk mitigation strategies for parts at risk by analysis and test
10	Monitor effectiveness of risk mitigation strategies during operations

variability within and between lots, and the acceptable risk. Typical design margins are factors up to 2 for rad-hard parts and up to 10 for COTS parts.

Additional references to support a radiation hardness assurance program are given in Table 9.7 with selected standards described next.

MIL-HDBK-814 addresses the piece-part level system engineering approach to the implementation of a radiation hardness assurance program applicable to both neutron and ionizing radiation permanent damage derived from nuclear weapon or natural space environments. A system of parts categorization based on design margin is identified that provides consistency in parts qualification procedures and control requirements. Information is provided on sampling and sample sizes, test facilities, qualification and lot acceptance testing, and documentation requirements.

MIL-HDBK-815 addresses dose-rate radiation effects on semiconductor electronics and hardness assurance at the piece-part level.

MIL-HDBK-816 provides guidelines and procedures for the preparation of detailed device specifications for the procurement of microcircuits and semiconductor devices where radiation hardness assurance is required.

MIL-HDBK-817 provides guidance to the project manager at both the sponsoring agency and the prime contractor. It assists the sponsoring agency in establishing survivability requirements and the needed contractual features in the

Table 9.7 Radiation hardness standards for electronics

Standard identifier	Title
MIL-HDBK-814 (U.S. DOD 1994a)	Ionizing Dose and Neutron Hardness Assurance Guidelines for Microcircuits and semiconductor Devices
MIL-HDBK-815 (U.S. DOD 1994b)	Dose-Rate Hardness Assurance Guidelines
MIL-HDBK-816 (U.S. DOD 1994c)	Guidelines for Developing Radiation Hardness Assurance Device Specifications
MIL-HDBK-817 (U.S. DOD 1994d)	System Development Radiation Hardness Assurance
MIL-STD-750 (U.S. DOD 1995)	Test Method Standard for Semiconductor Devices
MIL-STD-883F (U.S. DOD 2006)	Test Method Standard, Microcircuits
ASTM-F 1892-06 (ASTM 2006b)	Standard Guide for Ionizing Radiation (Total Dose) Effects Testing of Semiconductor Devices
ASTM-F 1893-98(2003) (ASTM 2003d)	Standard Guide for Measurement of Ionizing Dose-Rate Induced Burnout of Semiconductor Devices
ASTM-F 1192-00(2006) (ASTM 2006a)	Standard Guide for the Measurement of Single Event Phenomena (SEP) Induced by Heavy Ion Irradiation of Semiconductor Devices
ASTM-E 1249-00(2005) (ASTM 2005b)	Standard Practice for Minimizing Dosimetry Errors in Radiation Hardness Testing of Silicon Electronic Devices Using Co-60 Sources
ASTM F 744M-97(2003) (ASTM 2003a)	Standard Test Method for Measuring Dose Rate Threshold for Upset of Digital Integrated Circuits [Metric]
ASTM F 773M-96(2003) (ASTM 2003b)	Standard Practice for Measuring Dose Rate Response of Linear Integrated Circuits [Metric]
ASTM F 980M-96(2003) (ASTM 2003c)	Standard Guide for Measurement of Rapid Annealing of Neutron-Induced Displacement Damage in Silicon Semiconductor Devices [Metric]
ASTM F 1190M-99(2005) (ASTM 2005a)	Standard Guide for Neutron Irradiation of Unbiased Electronic Components
ASTM F 1262M-95(2002) (ASTM 2002)	Standard Guide for Transient Radiation Upset Threshold of Digital Integrated Circuits
ESA/SCC-22900 (ESA/SCC 1995)	Total Dose Steady-State Irradiation Test Method
ESA/SCC-25100 (ESA/SCC 1995)	Single Event Effects Test Method and Guidelines
EIA/JESD-57 (EIA/JESD 1996)	Test Procedures for the Measurement of Single-Event Effects in Semiconductor Devices from Heavy Ion Irradiation

Request for Proposal (RFP), the Statement of Work (SOW), and in identifying Data Item Descriptions (DIDs) for the Contract Data Requirements Lists (CDRLs). The prime contractor is aided by guidance in the structure and timeline of a Hardness Assurance Program, and the tasks needed as the development of the system progresses through design, development, and production phases.

MIL-STD-750D establishes uniform methods for testing semiconductor devices, including basic environmental tests to determine resistance to deleterious effects of natural elements and conditions surrounding military operations, and physical and electrical tests. The term devices includes items such as transistors, diodes, voltage regulators, rectifiers, tunnel diodes, and other related parts categorized as semiconductors.

MIL-STD-883F establishes uniform methods, controls, and procedures for testing microelectronic devices suitable for use in aerospace electronic systems, including basic environmental tests to determine resistance to deleterious effects of natural elements and conditions surrounding space operations; mechanical and electrical tests; workmanship and training procedures. The term devices includes such items as monolithic, multichip, film and hybrid microcircuits, microcircuit arrays, and the elements from which the circuits and arrays are formed that are categorized as microelectronic devices.

MIL-STD 750 and 883F consist of a plethora of individual tests of which a sample is given in Table 9.8.

ASTM F1892-04 gives the guidelines for establishing an appropriate sequence of tests and data analysis procedures for determining the ionizing radiation (total dose) hardness of microelectronic devices for dose rates below 300 rad (SiO₂)/s. These tests and analyses help in the determination of the capability of the devices to meet specific hardness requirements or to evaluate the parts for use in a range of radiation environments.

ASTM F1893-98 defines detailed requirements for testing microcircuits for short-pulse high-dose-rate ionization-induced failures. Large flash x-ray (FXR) machines operated in the photon mode, or FXR e-beam facilities, are required because of the high-dose-rate levels that are necessary to cause burnout. Two modes of test are possible: survival test and a failure test.

ASTM F1192-00 defines requirements and procedures for testing integrated circuits and other devices for the effects of single-event phenomena (SEP) induced by irradiation with heavy ions having an atomic number $Z \geq 2$. This includes any manifestation of upset induced by a single ion strike, including soft errors (one or more simultaneous reversible bit flips), hard errors (irreversible bit flips), latchup (permanent high conducting state), transients induced in combinatorial devices that may introduce a soft error in nearby circuits, and power field effect transistor (FET) burn-out and gate rupture.

Table 9.8 MIL-STD 883F-2004 radiation testing

Methods	Description
1017.2	Neutron irradiation
1019.6	Ionizing radiation (total dose) test procedure
1020.1	Dose rate induced latchup test procedure
1021.2	Dose rate upset testing of digital microcircuits
1022	MOSFET threshold voltage
1023.2	Dose rate response of linear microcircuits
1032.1	Package induced soft error test procedure (due to alpha particles)

Table 9.9 Radiation hardness assurance (RHA) categories

Radiation level designator	Radiation limit, krad (Si)
M	3
D	10
P	30
L	50
R	100
F	300
G	500
H	1000

Table 9.9 gives the letter designators used on components to identify the maximum radiation limits.

9.8 Radiobiology

9.8.1 Introduction

NASA's Bioastronautics Roadmap (Charles 2005) identified radiation and synergistic effects of radiation as potentially placing astronauts at significant risk for acute and long-term radiation sickness, such that space exploration missions or crew survival may be placed in jeopardy. Exposure to ionizing radiation encountered during space exploration may cause a number of health-related problems, but the primary concern is related to the increased risk of carcinogenesis in astronauts. A number of parameters affect astronaut exposure to radiation, including the structure of the spacecraft; the materials used to construct the spacecraft; the trajectory; flux density of the galactic cosmic rays, trapped radiation, and solar particles; phase of the solar cycle; and whether protection is afforded by the geomagnetic field. All of these must be taken into account to ensure that radiation exposures received by the astronauts remain below established safety limits.

Radiobiology is the branch of biology that specializes in the study of the effects of radiation on biological molecules, cells, tissues, and whole organisms including humans. Radiobiology seeks to understand the molecular changes responsible for radiation effects such as carcinogenesis, genetic changes, and cell impairment or death. Because of the intrinsic and methodological difficulties in determining if the health effects demonstrated at high radiation doses are also present at low doses, current radiation protection standards and practices are based on the premise that any radiation dose, no matter how small, may result in detrimental health effects.

Not all kinds of radiation are harmful to biological tissue. Non-ionizing radiation such as radio waves, microwaves, or infrared is only able to transfer heat to tissue. Consequently, the risk depends on the power density of the radiation. If the power is sufficiently low to produce no thermal damage, it is considered harmless. The effects of radiation on tissue are characterized as either deterministic (nonstochastic) or stochastic.

Deterministic radiation effects lead to a loss of function of the irradiated organ or tissue because of the death or impairment of a sufficient number of cells. Because the degree of the pathology (disease, loss, or damage) is proportional to the number of cells affected, a threshold must be exceeded for the loss of function to occur. For radiation doses above the threshold, the severity of the effect increases. Many noncancerous health effects of radiation are deterministic including erythema (skin reddening), cataracts (opacity of the eye lens), temporary and permanent sterility, and radiation sickness (nausea, diarrhea, headache, lethargy, fever, and possibly death).

Stochastic radiation effects of ionizing radiation are the occurrence of pathology whose probability of occurrence, but not severity, is a function of the dose without the existence of a threshold. The probability that stochastic radiation pathologies will occur differs widely for the individually irradiated organs or tissues. The most important stochastic effects, cancer and inheritable genetic damage, may appear many years or decades after exposure. It is thought that there is no minimum threshold for these effects; as the dose decreases, the effects are still expected to occur, but with lower frequency. Consequently, there is no firm biological basis for setting a safe level of exposure above the background for stochastic effects.

9.8.2 Radiobiology Units

Radiobiology units used in the interactions between radiation and tissue are summarized in Table 9.10 and described in the following text. These are exposure, absorbed dose, equivalent dose, dose equivalent, and effective dose.

9.8.2.1 Exposure. The *roentgen* (R) is a measure of exposure defined as the amount of charge induced in dry air by x-ray and gamma-ray radiation, where

Table 9.10 Radiation units and conversions

Quantity	SI units	Common units	Conversions
Exposure	Roentgen, R 1R of x rays or γ rays produce 2.58×10^{-4} coulombs of charge per kilogram of dry air	—	—
Absorbed dose	Gray, Gy Radiation energy absorbed in 1 kg of material	Radiation absorbed dose, rad Radiation energy absorbed in units of 100 ergs g^{-1} of material	1 rad = 0.01 Gy
Equivalent dose	Sievert, Sv	Roentgen equivalent man, rem	1 rem =
Effective dose	Relates absorbed dose to effective biological damage	Relates absorbed dose to effective biological damage	0.01 Sv

one roentgen produces 2.58×10^4 coulombs of charge per kilogram of dry air. The roentgen is approximately equal to 0.0093 Gy.

9.8.2.2 Absorbed dose. The *absorbed dose* D is the amount of any type of radiation energy deposited in a unit mass of any material. The SI unit is the gray (Gy), equivalent to a joule of energy deposited in 1 kg of material, J kg^{-1} . The common unit is the *radiation absorbed dose* (rad) that is equal to 100 ergs of energy deposited in a gram of material, erg g^{-1} . The equivalence is that $1 \text{ rad} = 0.01 \text{ Gy} = 0.01 \text{ J kg}^{-1}$.

9.8.2.3 Equivalent dose. The *equivalent dose* H_T is an attempt to provide a means to normalize the damage to human tissue when exposed to different types of radiation. For the same absorbed dose, the same or different types of radiation at different energies can produce different levels of damage. The equivalent dose relates the absorbed dose in human tissue to the *effective biological damage* caused by x-rays or gamma rays. The SI unit is the sievert (Sv), obtained by multiplying the absorbed dose in grays by a nondimensional radiation weighting factor. Consequently, equivalent dose has the same fundamental units as absorbed dose, J kg^{-1} , but is identified as sieverts to distinguish the absorbed from the equivalent dose. The common unit is the *roentgen equivalent man* (rem), which is equal to the radiation weighting factor times the absorbed dose in rads, with fundamental units erg g^{-1} . The equivalence is that $1 \text{ rem} = 0.01 \text{ Sv}$.

Mathematically the equivalent dose is based on an average absorbed dose in a tissue or an organ, D_T , and weighted by the radiation weighting factor, W_R , for the radiation(s) absorbed by the body or, in the case of internal emitters, the radiation emitted by the source. Therefore, the equivalent dose is the product of the average-absorbed dose, $D_{T,R}$, in a tissue or organ, T , due to radiation R and a radiation factor, W_R , for each radiation given by

$$H_{T,R} = W_R D_{T,R} \quad (9.34)$$

where

$H_{T,R}$ = equivalent dose in tissue type T from radiation type R , Sv

$D_{T,R}$ = absorbed dose in tissue type T from radiation type R , Gy

W_R = radiation weighting factor for radiation type R , dimensionless

Often the subscript T is not used if the calculation is for full-body radiation or the tissue is not differentiated. The radiation weighting factors, W_R , given in Table 9.11 are set by the International Commission on Radiological Protection (ICRP) in ICRP-60 (1991).

When the radiation field consists of mixed radiation, the total equivalent dose, H_T , is given by the sum of the equivalent doses for each type of radiation from ICRP-60 (International Commission on Radiological Protection 1991) as

$$H_T = \sum_R H_{T,R} = \sum_R W_R D_{T,R} \quad (9.35)$$

where

H_T = equivalent dose in tissue type T , Sv

Table 9.11 Radiation weighting factors

Radiation type and energy range	Radiation weighting factor, W_R	
	ICRP 60	ICRP 92
Photons, all energies	1	1
Electrons, positrons, and muons, ^a all energies	1	1
Neutrons		
<10 keV	5	A continuous curve is recommended with minimums of 5 at 0.001 MeV and 1000 MeV and a maximum of 20 at 0.5 MeV
10 keV to ≤100 keV	10	
>100 keV to ≤2 MeV	20	
>2 MeV to ≤20 MeV	10	
>20 MeV	5	
Protons (other than recoil protons) and energies >2 MeV	2–5	2
Alpha particles, fission fragments, heavy nuclei	20	20

^aMuon = mu mason, which is a short-lived negatively charged particle.

Source: From International Commission on Radiological Protection, ICRP 60 (1991) and ICRP 92 (2004).

Example 9.6

Problem: Determine the equivalent dose for tissue exposed to 5 Gy of gamma rays and 1 Gy of 1-MeV neutrons.

Solution: From Table 9.11, it follows that the radiation weighting factors are $W_{\text{gamma}} = 1$ and $W_{\text{neutrons at 1MeV}} = 20$, so that Eq. (9.35) gives

$$H = \sum_R W_R D_R = 1 \times 5 + 20 \times 1 = 25 \text{ Sv}$$

In this case, the neutrons dominate the tissue damage even though their absorbed dose is less.

9.8.2.4 Dose equivalent. The *dose equivalent* H is a measure of the absorbed dose at a point in tissue. Consequently, the dose equivalent is conceptually different from equivalent dose. The quality factor Q was introduced to define dose equivalent based on the fact that the probability of stochastic radiation effects is a function of the linear energy transfer, LET . The radiation quality factor Q as a function of LET in water is given in NCRP-132 (National Council on Radiation Protection and Measurements 2000) by

$$\begin{aligned} Q &= 1.0 & LET < 10 \text{ keV } \mu\text{m}^{-1} \\ Q &= 0.32 \text{ } LET - 2.2 & 10 \leq LET < 100 \text{ keV } \mu\text{m}^{-1} \end{aligned}$$

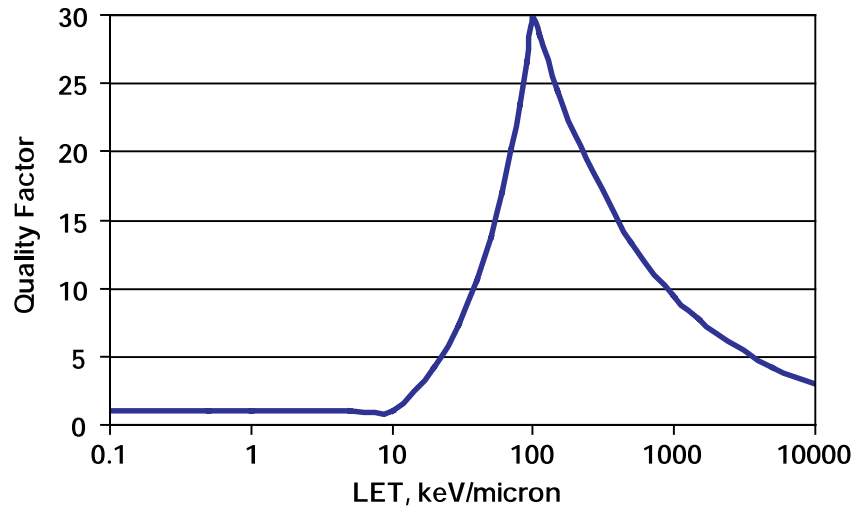


Fig. 9.22 Radiation quality factor as function of LET in water.

$$Q = 300 \, LET^{-1/2} \quad 10 \geq LET = 100 \, \text{keV } \mu\text{m}^{-1} \quad (9.36)$$

and is plotted in Fig. 9.22.

The dose equivalent H is given in terms of the radiation quality factor Q by

$$H = \int Q(L) D(L) \, dL \quad (9.37)$$

where

H = dose equivalent of the particular radiation, Sv

$D(L)$ = absorbed dose from radiation type R , Gy

$Q(L)$ = quality factor for radiation type R , dimensionless

$L \equiv LET$, linear energy transfer, $\text{keV } \mu\text{m}^{-1}$

9.8.2.5 Effective dose. *Effective dose* E is a means to assess quantitatively situations where the radiation is spatially non-uniform so that different parts of the body receive different radiation intensities. This would be the case if the shielding or the radiation is not spatially uniform as in radiation therapy or the ingestion of a radionuclide in a nuclear medicine test. The effective dose is the sum of the weighted equivalent doses for all irradiated tissues or organs. The *tissue weighting factor*, W_T , takes into account the level of radiation absorbed by each organ and tissue. Thus, tissue weighting factor represents the relative contribution of that organ or tissue to the total detriment that would result from uniform irradiation of the whole body. The tissue weighting factor includes the mortality and morbidity risks from cancer, risk of hereditary effects for all generations, and the length of life lost. Tissue weighting factors in Table 9.12 are values recommended by ICRP-26, ICRP-60, and ICRP-2005 (International Commission on Radiological Protection 1977, 1991, 2005); the latter has not yet been officially adopted.

Table 9.12 Tissue weighting factors W_T

Organ (tissue)	ICRP-26	ICRP-60	ICRP-2005
Gonads	0.25	0.20	0.05
Colon		0.12	0.12
Bone marrow (red)	0.12	0.12	0.12
Lung	0.12	0.12	0.12
Stomach		0.12	0.12
Bladder		0.05	0.05
Breast	0.15	0.05	0.12
Liver		0.05	0.05
Thyroid gland	0.03	0.05	0.05
Esophagus		0.05	0.05
Skin		0.01	0.01
Bone surface	0.03	0.01	0.01
Brain			0.01
Kidneys			0.01
Salivary glands			0.01
Remainder (adrenals, brain, upper large intestine, small intestine, kidney, muscle, pancreas, spleen, thymus, and uterus)	0.30	0.05	0.10
TOTAL	1.00	1.00	1.00

Source: From International Commission on Radiological Protection (2005).

The effective dose E in sieverts is determined from the summation of the equivalent dose in each tissue or organ multiplied by the weighting factor for that tissue or organ as

$$E = \sum_T W_T H_T = \sum_T W_T \sum_R W_R D_{T,R} \quad (9.38)$$

where

E = effective dose, Sv

H_T = equivalent dose in particular tissue or organ, Sv

W_T = tissue weighting factor, dimensionless

Consequently, if the whole body is uniformly exposed to the radiation, the effective dose is equal to the equivalent dose.

Example 9.7

Problem: Determine the effective dose if a subject's thyroid is exposed to an equivalent dose of 0.3 Sv.

Solution: From Table 9.12, the radiation weighting factor for the thyroid is $W_T = 0.05$. It then follows from Eq. (9.38) that for an equivalent dose $H_T = 0.3$ Sv that

$$E = \sum_T W_T H_T = 0.05 \times 0.3 = 0.015 \text{ Sv} \quad (9.39)$$

Thus, an equivalent dose of 0.3 Sv to the thyroid is equivalent to an effective dose of 0.015 Sv to the whole body.

9.8.3 Radiation Standards

The National Council on Radiation Protection and Measurements (NCRP) is a nonprofit organization chartered by the U.S. Congress that makes recommendations regarding protection against radiation. The International Commission on Radiological Protection (ICRP) is an independent registered charity established to advance for the public benefit the science of radiological protection, in particular by providing recommendations and guidance on all aspects of protection against ionizing radiation. These two organizations recommended acceptable levels of radiation exposure in publications, e.g., ICRP-60 (International Commission on Radiological Protection 1991) and NCRP-116 (National Council on Radiation Protection and Measurements 1993) for the general public and radiation workers. Astronauts are classified as radiation workers. The bases for the estimates are the cancer mortality experience for the survivors of Hiroshima and Nagasaki up to 1988 and other accidental exposures. Their recommendations are summarized in Table 9.13. A dose of 20 mSv per year is estimated to result in the death of one person in a thousand from radiation.

The NCRP has also issued two reports that provide recommendations on radiation protection guidance and radiation exposure limits for astronauts working in low Earth orbit, NCRP-98 issued in 1989, and the updated NCRP-132 [National Council on Radiation Protection and Measurements (2000)] in 2000. The deterministic limits of both sets of recommendations are given in Table 9.14. The unit of gray-equivalent (Gy-Eq), introduced in NCRP-132, is defined as the organ dose in grays multiplied by the relative biological effectiveness (RBE) for the specific organ and radiation. This is a change from the prior use of existing radiation quality factors based on data for late effects, most importantly cancer, that is a stochastic effect.

Career dose limits for stochastic effects for exposure in low Earth orbits are given in Table 9.15. These limits correspond to a probability of 3% excess cancer mortality for a 10-year career and are dependent on age and gender. In the more recent guidance in NCRP-132, the exposure limits are reduced, based on findings from more recent analyses of the survivors of the atomic bomb events during World War II and other exposed cohorts.

Effects from exposure to radiation are classified as either acute or delayed. *Acute effects* occur within two months of exposure and are deterministic. *Delayed*

Table 9.13 Radiation dose limits

Radiation exposure	ICRP-60 (1991)	NCRP-116 (1993)
Radiation Workers	50 mSv annual effective dose limit and 100 mSv in 5 yr cumulative effective dose limit	50 mSv annual effective dose limit and 10 mSv $\times$ age (yr) cumulative effective dose limit
Stochastic effects		
Deterministic effects	150 mSv equivalent dose limit to lens of eye and 500 mSv annual equivalent dose limit to skin, hands, and feet	150 mSv annual equivalent dose to lens of eye and 500 mSv annual equivalent dose limit to skin, hands, and feet
General public		
Stochastic effects	1 mSv annual effective dose limit, higher values acceptable if the annual average over 5 yr does not exceed 1 mSv	1 mSv annual effective dose limit for continuous exposure, and 5 mSv annual effective dose limit for infrequent exposure
Deterministic effects	15 mSv annual effective dose limit to lens of eye and 50 mSv annual equivalent dose limit to skin	50 mSv annual equivalent dose limit to skin and extremities

Source: From ICRP-60 (International Commission on Radiological Protection 1991) and NCRP-116 (National Council on Radiation Protection and Measurements 1993).

Table 9.14 Radiation dose limits for deterministic effects for exposure in low Earth orbits

Anatomical location	NCRP-98 (NCRP 1989), Sv			NCRP-132 (NCRP 2000) Gy-Eq		
	30-day limit	1-year limit 2.0	Career	30-day limit	1-year limit	Career
Eyes	1.0	2.0	4.0	1.0	2.0	4.0
Skin	1.5	3.0	6.0	1.5	3.0	6.0
Blood forming organs	0.25	0.5				
Bone marrow				0.25	0.5	

Table 9.15 Career radiation dose limits for stochastic effects for exposure in low Earth orbits, based on excess risk of cancer mortality of 3%

Age (yr)	Career exposure limits, NCRP-98 (NCRP 1989), Sv		10-year career exposure limits, NCRP-132 (NCRP 2000), Sv	
	Male	Female	Male	Female
25	1.5	1.0	0.7	0.4
35	2.5	1.75	1.0	0.6
45	3.25	2.5	1.5	0.9
55	4.0	3.0	3.0	1.7

Table 9.16 Radiation risk values

Radiation risk	ICRP-60 (ICRP 1991), Sv ⁻¹	NCRP-116 (NCRP 1993), Sv ⁻¹
Adult workers	4.0×10^{-2} for fatal cancer 0.8×10^{-2} for nonfatal cancer 0.8×10^{-2} for severe genetic effects	4.0×10^{-2} for fatal cancer 0.8×10^{-2} for nonfatal cancer 0.8×10^{-2} for severe genetic effects
Whole populations	5.0×10^{-2} for fatal cancer 1.0×10^{-2} for nonfatal cancer 1.3×10^{-2} for severe genetic effects	5.0×10^{-2} for fatal cancer 1.0×10^{-2} for nonfatal cancer 1.3×10^{-2} for severe genetic effects
Embryo-fetus		$\sim 10 \times 10^{-2}$

effects occur at two or more months after exposure and can be either stochastic or deterministic. An assessment of the stochastic risk of fatal cancer, nonfatal cancer, and severe genetic effects from radiation exposure are given in Table 9.16. The risk values describe the (assumed stochastic) relationship between the estimated dose incurred by an individual and the expectation value of his/her combined risk of harmful effects from exposure in units of risk per Sv.

Ionizing radiation can cause deterministic radiation sickness that may have symptoms of nausea and vomiting; diarrhea; erythema (skin redness); weakness; fatigue; exhaustion; fainting; dehydration; tenderness; swelling; internal bleeding; hair loss; ulceration of the mouth, esophagus, stomach, or intestines; vomiting of blood; bleeding from the mouth, gums, and rectum; bruising, peeling of skin; coma; and possibly death. Symptoms as a function of the equivalent dose (dose x-rays or gamma rays is the same as equivalent dose) to which a subject is exposed are presented in Table 9.17. LD₅₀ is a statistically derived single dose of an agent (in our case radiation) that can be expected to cause death in 50% of a given population of organisms (humans) under a defined set of experimental conditions.

Table 9.18 illustrates the sources of exposure to ionizing radiation in the United States.

Table 9.17 Health effects due to radiation exposure

Health effect	Dose, x or gamma radiation, Gy
Blood count changes in population	0.15–0.25
Blood count changes in individual	0.5
Vomiting, effective threshold	1.0
Mortality, effective threshold	1.5
LD ₅₀ with minimal supportive care	3.2–3.6
LD ₅₀ with supportive medical treatment	4.8–5.4
LD ₅₀ with autologous bone marrow or blood stem cell transplant	11.0

Source: From NCRP-98 (National Council on Radiation Protection and Measurements 1989); reprinted with permission of the National Council on Radiation Protection and Measurements.

Table 9.18 Radiation exposure in the United States, average of 2.8 mSv

Source	Equivalent or effective dose, mSv	
Natural		
Radon	2.0	
Other than radon	1.0	
Subtotal		3.0
Nuclear fuel cycle		0.0005
Consumer products ^a		0.09
Medical		
Diagnostic x-rays	0.39	
Nuclear medicine	0.14	
Subtotal		0.53
Total	3.5	3.6

^a Building materials, television, luminous watches, smoke detectors, etc.

Source: From NCRP-93 (National Council on Radiation Protection and Measurements, 1987).

9.8.4 Radiation Risk During Spaceflight

Prohibitive shielding with areal densities $\sim 300\text{--}500\text{ g cm}^{-2}$ would be necessary to protect astronauts if their radiation limits were set at levels comparable to those of occupationally exposed individuals on Earth or at the even lower exposure limits established for the general public (National Academy of Sciences 2006). The typical spacesuit has a shielding of $\sim 0.19\text{ g cm}^{-2}$, the space shuttle has shielding of $\sim 10\text{ g cm}^{-2}$, and the ISS has maximum shielding of $\sim 15\text{ g cm}^{-2}$. Future outposts on the moon or Mars will have storm shelters, most likely with polyethylene and aluminum possibly exceeding just 20 g cm^{-2} . Astronaut limits for operations in low Earth orbit (LEO) are approximately an order of magnitude higher than limits for earthbound radiation workers, 50 cSv per year for astronauts, with a lifetime limit that depends on age and sex; however, no limits have been established as yet for lunar or Mars missions. Difference in allowed exposure rates is due to the shorter career exposure times for astronauts (assumed <10 years) vs possible >40 years for radiation workers. The LEO limits for astronauts, although higher than limits for earthbound radiation workers, are based on a 3% excess cancer mortality risk. Shielding needed to attain this elevated level of permitted exposure is in the range of $>20\text{ g cm}^{-2}$. For 20 g cm^{-2} aluminum shielding, Townsend et al. (1992) estimate 50 cSv y^{-1} at solar minimum, but Cucinotta et al. (2005) now estimate closer to 75 cSv y^{-1} using newer transport codes and environmental models.

An astronaut is exposed to radiation on Earth, during ascent into orbit, during spaceflight, while deorbiting, and on arrival back on Earth. The total cumulative dose is the most important criterion for determining astronaut risk. The extreme solar particle event of 4 August 1972 has been studied with regard to astronaut safety (Wilson et al. 1997). Figure 9.23 gives the fluence of the August 1972 solar particle event that reached a peak of $20 \times 10^{10}\text{ protons cm}^{-2}$ on 5 August.

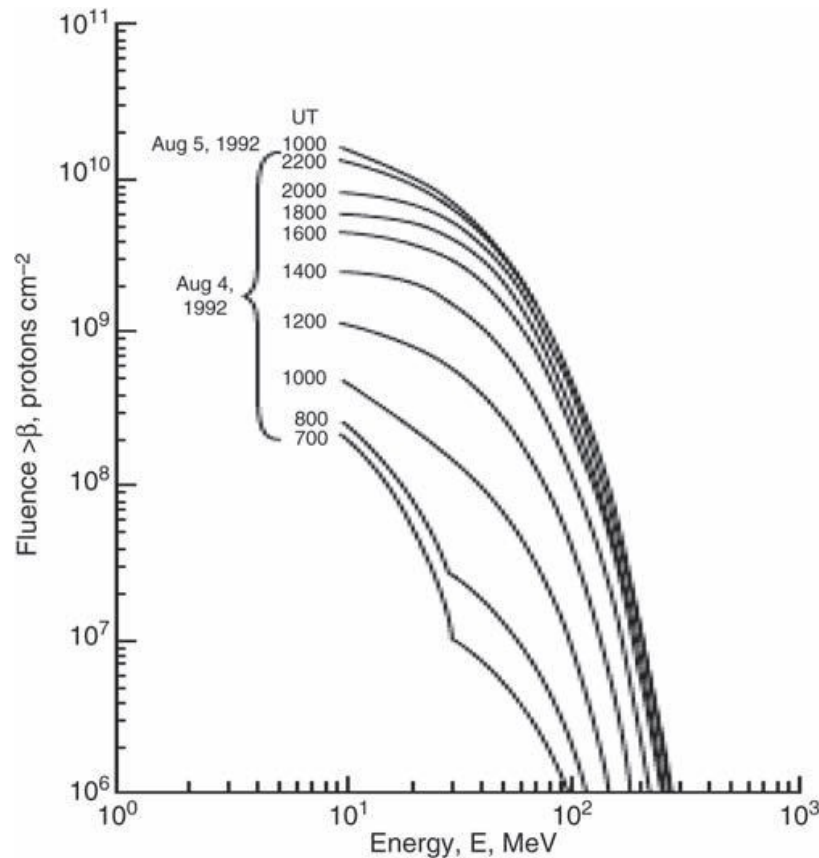


Fig. 9.23 Radiation fluence of August 1972 solar particle event. [From Wilson et al. (1997); courtesy of NASA.]

Figure 9.24a gives the risk of cancer for a given dose as a function of exposure time. It is parameterized for a dose of 1 and 2 Gy as well as the exposure predicted to be received by an astronaut in a space suit or in a pressure vehicle during the August 1972 event. For the same dose, as the exposure time increases, the risk is reduced. Figure 9.24b gives the predicted percent mortality as a function of dose rate for a total dose of 2 Gy for an astronaut either medically untreated or medically treated. Also shown is the predicted mortality for an astronaut either in a space suit or in a pressure vehicle during the August 1972 event and at different rates of exposure.

Figure 9.25 gives the recorded history of radiation exposure of astronauts as recorded by their personal dosimeters. The larger values are characterized by longer durations in space and higher inclinations as on the MIR at 51.6-deg inclination.

Table 9.19 gives the average dose D or dose rate recorded by dosimeter badges, and estimates of the dose equivalent H to the blood forming organs (BFO) received by crews in NASA missions through 1999.

Table 9.20 gives the fatal cancer projections and 95% confidence interval (CI) for 40-year-old females for several exploration missions. Aluminum shielding of 4 g cm^{-2} is assumed, along with a Martian atmosphere for the Mars scenarios. Also illustrated is the effect of additional shielding of 10 cm of water. Table 9.21

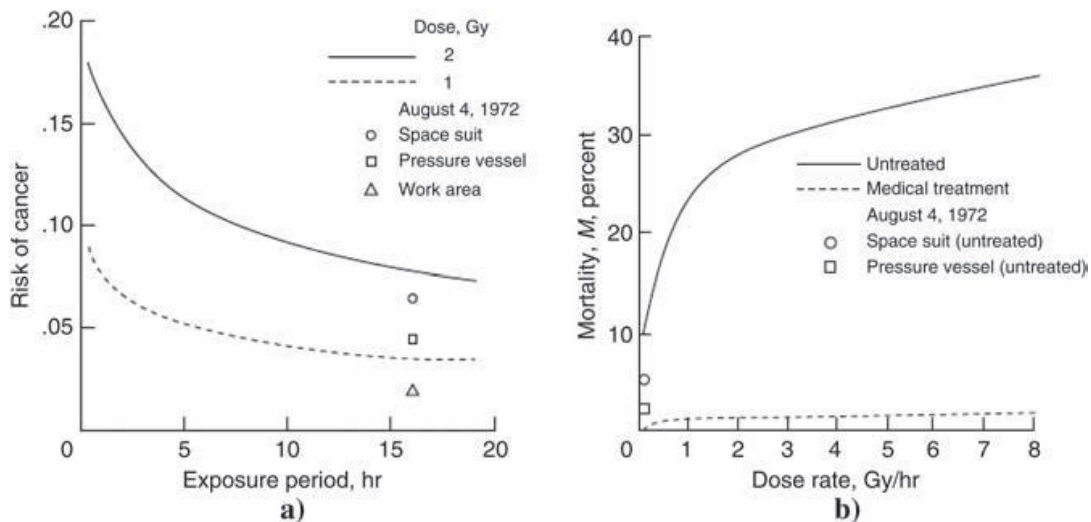


Fig. 9.24 Risk of cancer and mortality for 4 August 1972 solar particle event: a) cancer risk for 1 and 2 Gy exposure, b) mortality for 2 Gy bone marrow dose. [From Wilson et al. (1997); courtesy of NASA.]

gives a similar result for a 40-year-old male. In each case, the uncertainties, indicated by the brackets, are exceptionally large.

Table 9.22 illustrates the maximum mission days in deep space for a 95% CI to remain below a 3% risk of excess cancer fatality. Shielding by the body and aluminum of 10 g cm^{-2} are assumed. The calculations are for solar minimum when the flux density of the galactic cosmic rays is a maximum.

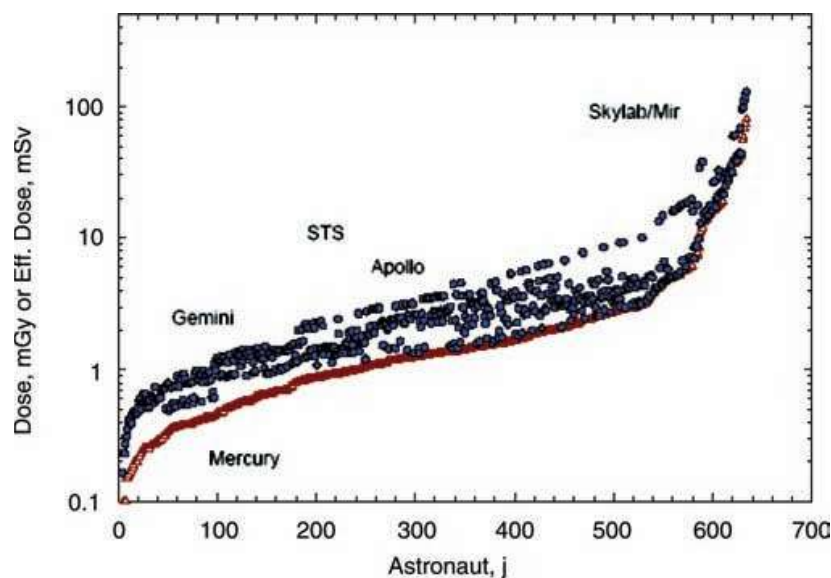


Fig. 9.25 Historical radiation doses recorded by astronauts ▲ and estimates of effective doses ● for all NASA missions through December 1999. [From Cucinotta et al. (2002); courtesy of NASA.]

Table 9.19 Average measured dose D and dose rate, and estimates of dose equivalent H and dose equivalent rates to BFO for NASA missions through 1999

NASA program	Inclination, deg	Altitude, km	Crew, number	D , mGy	H , mSv	D rate, mGy/day	H rate mSv/day
Mercury	—	—	6	0.1	0.15	0.3	0.55
Gemini	—	—	20	1.3	2.2	0.49	0.87
Apollo	—	—	33	4.1	12.0	0.43	1.2
Skylab	50	430	9	40.3	95.0	0.71	1.40
ASTP	50	220	3	1.1	2.3	0.12	0.26
STS	28.5	>400	85	9.5	17.0	1.2	2.1
SIS	28.5	<400	207	0.9	1.6	0.1	0.18
SIS	39–40	~400	57	1.1	2.4	0.1	0.21
STS	>50	>400	10	2.2	5.2	0.44	1.1
SIS	>50	<400	190	1.7	3.8	0.2	0.45
NASA-Mir	51.6	~390	7	50.3	115	0.37	0.84

Source: From Cucinotta et al. (2002); courtesy of NASA.

Table 9.20 Fatal cancer risk projections with 95% confidence interval for 40-year-old females for several exploration missions with 4 g cm^{-2} shielding of aluminum, Mars atmosphere, and effect of shielding of 10 cm of water

Mission type	Total duration in days (on Mars or lunar surface)	%Probability of excess fatal cancer	
		0 cm H ₂ O %	10 cm H ₂ O
Deep space	62 (0)	0.6 [0, 3.3]	0.45 [0, 2.7]
Lunar base	20 (14)	0.13 [0, 0.7]	0.09 [0, 0.6]
Mars-1	360 (30)	3.3 [0, 18.0]	2.5 [0, 14.6]
Mars-2	660 (30)	6.2 [0, 34.0]	4.6 [0, 27.5]
Mars-3	1000 (600)	5.7 [0, 30.8]	4.5 [0, 25.6]

Source: From Cucinotta et al. (2002); courtesy of NASA.

Table 9.21 Fatal cancer risk projections with 95% confidence interval for 40-year-old males for several exploration missions with 4 g cm^{-2} shielding of aluminum, Mars atmosphere, and effect of shielding of 10 cm of water

Mission type	Total duration in days (on Mars or lunar surface)	%Probability of excess fatal cancer	
		0 cm H ₂ O	10 cm H ₂ O
Deep Space	62 (0)	0.4 [0, 2.0]	0.27 [0, 1.6]
Lunar Base	20 (14)	0.08 [0, 0.41]	0.06 [0, 0.34]
Mars-1	360 (30)	2.0 [0, 10.8]	1.5 [0, 8.8]
Mars-2	660 (30)	3.7 [0, 20.4]	2.8 [0, 16.5]
Mars-3	1000 (600)	3.4 [0, 18.5]	2.7 [0, 15.3]

Source: From Cucinotta et al. (2002); courtesy of NASA.

Table 9.22 Projections of maximum days in space for 95% confidence interval below probability of 3% excess fatal cancer

Age	Female, projected days	Male, projected days
30	54	91
35	62	104
40	73	122
45	89	148
50	115	191
55	159	268

Source: From Cucinotta et al. (2002); courtesy of NASA.

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Problems

- 9.1** Determine the thickness of aluminum that is required to reduce the flux density of a 200-keV gamma ray to 10% of its incident radiation. Assume the half-value layer for 200-keV gamma rays in aluminum is 2.14 cm.

- 9.2** Determine the fraction remaining of the flux density of 1-MeV gamma rays that travel 1 cm and 10 cm in lead and 10-MeV gamma rays that travel 1 cm and 20 cm in Lucite. Use the half-thickness layers for lead and Lucite given in Fig. 9.7.
- 9.3** (a) Derive the equation for the one-tenth-value layer (TVL) in terms of the attenuation coefficient.
(b) Derive the equation for the flux density in terms of the number n of one-tenth-value layers.
- 9.4** (a) Determine from the conservation of momentum and energy the velocity of the stationary particle after a head-on collision of two particles.
(b) Determine the energy of the stationary particle after the collision E_2' in terms of the energy of the moving particle prior to the collision E_1 .
(c) Show that the maximum energy is transferred when the particles have equal mass.
- 9.5** (a) Determine the charge transfer rate in pC μm^{-1} for a particle in silicon with an LET of 370 MeV $\text{cm}^2 \text{mg}^{-1}$.
(b) Repeat part (a) for gallium arsenide.
- 9.6** What is the stopping power required to deposit a charge of 0.1 pC μm^{-1} in silicon?
- 9.7** Determine the equivalent dose if a person were to receive an absorbed dose of 0.3 mGy of slow neutron radiation (<10 keV), 6 mGy of gamma radiation, and 0.1 mGy of fast neutron radiation (100 keV to 2 MeV).
- 9.8** Determine the effective dose if an individual has his lungs exposed to 2 mSv of radiation and his thyroid to 1 mSv.
- 9.9** A 5.0-kg mass of body tissue absorbs an average power of 30.0 W from a burst of x-rays lasting 0.50 s
(a) Determine the amount of energy absorbed from the beam.
(b) Determine the absorbed dose in the tissue.
(c) Determine the equivalent dose absorbed by the tissue.
- 9.10** A 5.0-kg mass of body tissue absorbs an average power of 30.0 W from a burst of alpha particles lasting 0.50 s.
(a) Determine the amount of energy absorbed from the beam.
(b) Determine the absorbed dose in the tissue.
(c) Determine the equivalent dose absorbed by the tissue.
- 9.11** A 5.0-kg mass of body tissue absorbs an average power of 30.0 W from a burst of x-rays lasting 0.50 s and at the same time absorbs an average power of 30.0 W from a burst of alpha particles lasting 0.50 s.
(a) Determine the amount of energy absorbed from the beam.
(b) Determine the absorbed dose in the tissue.